

More Trade, Less Diffusion: Technology Transfers and the Dynamic Effects of Import Liberalization*

Gustavo de Souza[†]

Ruben Gaetani[‡]

Martí Mestieri[§]

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Abstract

Long-run economic growth depends on the international diffusion of frontier technologies. Using Brazilian data, we identify a channel through which tariff cuts slow this diffusion: they weaken foreign firms' incentives to transfer technology to domestic producers. Exploiting variation in import tariffs across origin countries within narrowly defined industries, we find that tariff reductions lead to fewer technology transfers and fewer citations to foreign technology, with the largest declines occurring among firms located near previous technology recipients. To interpret these findings, we develop a growth model in which foreign firms choose between exporting goods and transferring technology, with learning from exports being less efficient than learning from transferred technologies, as informed by the empirical evidence. Trade liberalization shifts learning from transferred technologies to imported goods, raising welfare in the short run but slowing diffusion and productivity growth. An optimal subsidy to technology transfers substantially amplifies the welfare gains from trade liberalization.

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1 Introduction

A long-standing view in economic growth is that catch-up growth is primarily driven by the diffusion and adoption of existing frontier technologies rather than by domestic innovation (e.g., [Aghion et al., 2014](#); [Comin and Mestieri, 2014](#)). Consistent with this view, more open economies tend to grow faster, a pattern that has been interpreted as reflecting greater exposure to foreign technologies and ideas ([Lucas, 2009](#); [Lind and Ramondo, 2019](#)). Motivated by this observation, a growing body of work has formalized the link between openness and technology diffusion, emphasizing mechanisms operating through trade, firm interactions, and knowledge exchange (e.g., [Buera and Oberfield, 2020](#), [Perla et al., 2021](#), and [Sampson, 2016](#)). Despite this progress, direct evidence on the underlying mechanisms shaping diffusion remains limited.

This paper addresses this gap by providing direct evidence on the effect of trade policy on international technology diffusion. We study Brazil’s trade liberalization, using technology transfers—contracts through which foreign firms license production know-how to Brazilian producers—as our primary measure of technology diffusion. We show that, within narrowly defined industries, liberalization slowed diffusion from source countries that experienced larger tariff cuts: tariff reductions led to fewer technology transfers and fewer citations to foreign technology, with the steepest declines among firms located near earlier technology transfer recipients. We interpret this pattern as resulting from foreign firms’ choice between exporting final goods to Brazil and licensing the know-how needed to produce them locally. When tariffs fall, exporting becomes relatively more attractive, so firms substitute away from licensing. Because firms learn more from hands-on licensing relationships than from exposure to imported goods, this shift slows diffusion. A quantitative model calibrated to our estimated elasticities implies that subsidizing technology transfers is necessary to capture the full gains from trade liberalization.

We measure technology diffusion with data on technology transfers and patent citations. First, we collect comprehensive records on technology transfer contracts from the Brazilian patent office, capturing every instance in which foreign industrial knowledge—whether in the form of patents, technical assistance, or know-how—is licensed or sold to domestic Brazilian firms. These contracts are the primary channel through which foreign producers transfer cutting-edge technologies to local firms and have been shown to cause significant firm growth ([Giorcelli, 2019](#); [Giorcelli and Li, 2021](#); [Choi and Shim, 2023](#); [de Souza, 2020](#); [Choi et al., 2025](#)). Second, using PATSTAT and building on an extensive literature that uses citations to measure diffusion ([Prato, 2025](#); [Liang et al., 2024](#);

Ayerst et al., 2023; Ahn et al., 2026; Akcigit et al., 2024; Acemoglu et al., 2016; Büttner et al., 2022; Berkes et al., 2025), we track the citations that Brazilian patents make to foreign patents, measuring the extent to which Brazilian firms adopt or build upon foreign ideas. The combination of technology transfer and patent citation data provides a rich view of cross-border knowledge flows, capturing both deliberate transfers and broader patterns of idea diffusion leaving a paper trail in patented inventions.

We identify diffusion by comparing outcomes from different origin countries within narrowly defined industries. We estimate gravity equations of knowledge flows on tariffs with a complete set of fixed effects, allowing us to isolate diffusion from other channels impacted by trade liberalization, such as changes in market structure and competition. Therefore, our identifying variation comes from comparing the growth in knowledge flows across markets with different tariff changes, controlling for common shocks at the country and sector level.

To address the endogeneity of tariff changes and identify the causal effect of tariffs on knowledge flows, we adopt the instrumental variable strategy proposed by Boehm et al. (2023). The key idea behind the instrument is that changes in Most Favored Nation (MFN) tariffs are primarily driven by factors related to large MFN trade partners, creating plausibly exogenous variation in tariffs for small MFN partners. Extending Boehm et al. (2023), we validate the instrument by showing it is uncorrelated with (i) political connections and other major policies, (ii) origin-market productivity and growth, and (iii) innovation and citation activity in the origin markets. We also show that current tariffs have no effect on lagged outcomes, suggesting that the instrument is not picking up pre-existing trends.

We identify a channel through which trade liberalization slowed technology diffusion in Brazil: lower tariffs reduced foreign firms’ incentives to transfer technologies to local firms. First, lower tariffs reduce technology transfers: markets with a 10-percentage-point larger tariff decline received 1.27% fewer transfers than other markets. This decline is concentrated in transfers that include technological assistance and operates mainly on the extensive margin, as fewer foreign firms transfer and fewer domestic firms receive technology. Second, tariff cuts reduce Brazilian firms’ citations of foreign patents, with the largest declines among firms with limited exposure to foreign markets, such as non-importers and non-exporters. Third, using the technology proximity measure of Bloom et al. (2013), we find that tariff cuts increase the technological distance between Brazilian and foreign patents. Taken together, these results imply that greater trade openness can slow technology diffusion.

We further show that the declines in citations and technology transfers are interconnected. The drop in citations is most pronounced among firms located near recipients of technology transfers and is driven primarily by reduced citations to firms that previously transferred technology to Brazil. Using a causal forest approach (Wager and Athey, 2018), we show that country-sector pairs that more strongly reduce technology transfers in response to a drop in tariffs also experience more significant drops in citations. These findings indicate that reductions in technology transfers translate into a lower diffusion of foreign technologies among domestic firms, highlighting the importance of technology transfers as a channel of idea diffusion.

To understand whether diffusion through technology transfers is economically meaningful, we examine how tariffs affect citations made by high quality patents in Brazil. We measure quality in two ways. First, with the number of citations the patent received within three years of publication, a standard proxy for patent quality that predicts subsequent firm growth (Kortum, 1997; Kogan et al., 2017; Farre-Mensa et al., 2020). Second, we measure growth of the patenting firm over the three years after the patent is filed. Using both metrics, we find that tariff cuts reduce foreign citations most sharply for high-quality patents, suggesting a decline in high-quality innovation. Taken together, these patterns suggest that the slowdown in diffusion induced by the liberalization may have weakened aggregate growth, an implication we later evaluate through the quantitative model.

Additionally, we find that declining tariffs have no significant impact on the foreign ownership of Brazilian firms, ruling out foreign direct investment (FDI) as the primary driver of reduced knowledge diffusion.¹ Similarly, firms' connections to foreign suppliers and changes in import tariffs in the countries where technology transfers originate also do not appear to play a role. These findings suggest that lower tariffs slow the diffusion of foreign knowledge within the local economy primarily by reducing technology transfers, rather than through changes in ownership structures, supplier relationships, or external trade policies.

To move from the relative effects identified in the data to aggregate effects, we develop a multi-

¹This result is consistent with a large empirical literature showing null or weak effects of tariffs on FDI (Blonigen, 2005; Blonigen and Piger, 2014; Blonigen, 2002; Wheeler and Mody, 1992; Kinda, 2010; Moder and Spital, 2025; Matsuura and Hayakawa, 2012). Theory also implies an ambiguous effect: higher tariffs can encourage local affiliate production (tariff-jumping), but they can also raise the cost of operating affiliates when production relies on imported intermediates (Ramondo and Rodríguez-Clare, 2013; Helpman, 1985). Our results are consistent with these forces roughly canceling in vertical FDI, as in Blonigen and Piger (2014) and Matsuura and Hayakawa (2012). Other determinants, such as institutions and infrastructure, may have stronger effects on FDI than tariffs (Kinda, 2010; Matsuura and Hayakawa, 2012). Consistent with this view, the OECD points to infrastructure bottlenecks, high entry costs driven by institutional complexity, and weak institutions as key reasons for Brazil's low level of FDI (OECD, 2018). For a review of the literature, see Blonigen (2005).

country, multi-sector growth model in which, building on [Buera and Oberfield \(2020\)](#), contact of domestic firms with foreign technologies drives idea diffusion. Foreign firms that serve Brazil choose between exporting the final good or transferring their technology to a Brazilian producer. Because lower tariffs raise the profitability of exporting, tariff cuts reduce technology transfers, consistent with what we find in the data.

A key innovation of the model is that Brazilian firms acquire foreign knowledge through two channels: exposure to imported goods and interactions with local producers that have received technology transfer. The model allows learning efficiency to differ across these channels. Intuitively, and consistently with the empirical evidence, technology transfer is associated with more intensive learning through training, technical assistance, and the transmission of tacit knowledge, while exporting might provide limited opportunities for learning from foreign producers. For that reason, it is plausible that learning from technology transfer is more efficient than learning from interacting with foreign suppliers.

Two parameters determine how tariffs affect diffusion and growth: (i) the elasticity of technology transfer with respect to tariffs and (ii) the learning efficiency of technology transfer relative to learning from imports. The elasticity determines how sharply technology transfers fall when tariffs decline. Relative learning efficiency determines whether this shift away from transfers lowers or raises overall exposure to foreign knowledge. If technology transfer generates more learning than imports, tariff cuts reduce knowledge exposure by discouraging transfers. If imports generate more learning than transfers, tariff cuts expand learning through imported goods and increase diffusion. The model’s aggregate predictions therefore depend on the size of the transfer elasticity and on which channel—transfers or imports—delivers stronger learning.

We use the empirical estimates to calibrate these key parameters. We derive a set of gravity equations from the model similar to those estimated in the empirical analysis.² The model delivers closed-form expressions that map the estimated tariff effects on technology transfers and citations into the two key parameters: the elasticity of technology transfer with respect to tariffs and the learning efficiency of technology transfer relative to imports. In the data, a tariff increase increases both citations and technology transfers. That joint increase implies that learning through

²These gravity equations also clarify why our gravity structure is essential. A sector-level regression of tariffs on citations or technology transfers would confound diffusion with other tariff-driven forces, such as changes in competition and innovation incentives. By including sector-year fixed effects, the gravity specification absorbs these channels so the tariff coefficient isolates diffusion. The same equations also make the key identifying assumption transparent: the instrument must be orthogonal to foreign productivity shocks. Consistent with this assumption, we find no correlation between the instrument and patent growth or TFP growth in the foreign markets.

technology transfer is more efficient than learning through imported goods.

We find that accounting for technology transfers reverses the standard prediction that static welfare gains from trade are greater in the long run than in the short run. Tariff reductions weaken foreign producers’ incentives to transfer technology, inducing them to favor exports instead. While this raises welfare in the short run, it slows diffusion and reduces productivity growth in the recipient country. In our baseline experiment, a full trade liberalization in Brazil generates on average welfare gains of 0.27% in the long run, relative to a status quo in which tariffs do not change, while short-run gains are more than twice as large, at 0.69%.³

Because spillovers from technology transfers generate a dynamic externality, subsidies to technology transfers are necessary to fully realize the benefits of trade liberalization. We compute the optimal subsidy and show that, while it distorts the static allocation and reduces welfare in the short run, it substantially increases the number of technology transfers, thereby accelerating diffusion and boosting long-run productivity growth. The optimal subsidy balances these short-run losses against long-run gains. Implementing this policy increases the overall welfare gains from trade liberalization by a factor of five.

Related Literature. This paper contributes to a broad literature studying the role of knowledge diffusion in long-run economic growth and development. A central insight of this literature is that productivity growth depends not only on the creation of new ideas, but also on the speed and extent to which frontier technologies diffuse across firms, sectors, and countries (Sachs et al., 1995; Lucas, 2009; Lind and Ramondo, 2019). A growing body of work studies how policy-induced changes in economic integration shape the diffusion of knowledge across countries and, in turn, productivity growth. Quantitative models of growth in open economies emphasize that trade affects the set of foreign ideas to which domestic producers are exposed, as well as the intensity of learning from these sources (Buera and Oberfield, 2020; Hsieh et al., 2023). In these frameworks, trade policy influences not only static resource allocation, but also the dynamic evolution of productivity through its impact on international knowledge flows. Despite this progress, direct evidence on the specific firm-level mechanisms through which policy shapes international knowledge diffusion remains limited. Our work expands these frameworks by endogenizing firms’ decisions to transfer technology across borders and by quantifying how these decisions shape diffusion, growth, and

³By contrast, in a counterfactual exercise in which we shut down the technology-transfer margin and allow diffusion to happen only through exposure to imported goods, we find a substantial overestimate of welfare gains, which reach 5.98% per period under this assumption. Thus, accounting for diffusion through technology transfers reduces the dynamic overall welfare gains from trade liberalization by an order of magnitude.

welfare over time.

This paper is also closely related to the vast literature on technology diffusion and adoption in economic development. Our paper relates to recent work highlighting the role of industrial policy instruments—including trade policy, subsidies, and contractual arrangements—in shaping firms’ incentives to adopt and diffuse technology. For example, [Choi and Levchenko \(2021\)](#) and [Kim et al. \(2021\)](#) study how industrial policies affect firm productivity, innovation, and long-run growth in Korea, while other contributions emphasize the role of local development policies and institutional environments in facilitating technology adoption ([Giorcelli, 2019](#); [Giorcelli and Li, 2021](#); [Choi and Shim, 2023](#)).⁴

A related strand of work focuses on specific channels through which technology diffuses across borders, including technology transfer contracts, licensing agreements, and joint ventures. [Santacreu \(2025\)](#), [de Souza \(2020\)](#), and [Sampson \(2024\)](#) study the determinants and consequences of international technology transfer agreements, while [Choi et al. \(2025\)](#) analyze the role of joint ventures in facilitating cross-border diffusion. These studies highlight how firm-level incentives and contractual choices shape the direction and speed of knowledge flows. Our paper contributes to this literature by providing new empirical evidence on how import tariffs affect international knowledge diffusion through their impact on technology transfer contracts, and by embedding this mechanism in a quantitative framework that allows us to evaluate its dynamic welfare consequences.

Finally, our analysis relates to work examining how trade policy affects firms’ incentives for technology adoption, upgrading, and selection (e.g., [Bustos, 2011](#); [Sampson, 2016](#); [Farrokhi and Pellegrina, 2023](#); [Perla et al., 2021](#); [Bai et al., 2024](#)), as well as how technology embodied in traded inputs diffuses internationally ([Ayerst et al., 2023](#)). While these papers emphasize important channels through which openness affects productivity, we show that allowing technology transfer decisions to respond endogenously to trade policy can overturn standard predictions regarding the dynamic gains from trade. In particular, we demonstrate that trade liberalization may slow diffusion by reallocating learning toward less efficient channels, with first-order implications for long-run growth and welfare.

The remainder of the paper is organized as follows. Section 2 presents the data sources used in the analysis. Section 3 presents the empirical strategy and Section 4 the empirical results. Section 5 introduces the model, Section 6 derives the model-based gravity equations that allow us to map the

⁴In recent work, [Juhász et al. \(2024\)](#) provides a nuanced view of how industrial policies are influenced by political and economic constraints, emphasizing the importance of context-specific factors such as governance structures.

key parameters to the gravity coefficients, and Section 7 presents the calibration and quantitative experiments. Section 8 concludes.

2 Data

In this section, we describe a country-by-sector dataset that traces part of the knowledge flowing into each Brazilian sector from abroad. We use administrative records on international technology transfers to identify the origin of foreign technologies used by domestic firms. We complement these records with patent citation data from PATSTAT to infer which countries domestic firms are learning from. To test alternative mechanisms, we also use firm ownership data to measure foreign direct investment (FDI). We merge these measures with Brazilian import-tariff rates.

2.1 Technology Transfers

Our main dataset, collected by [de Souza \(2020\)](#), contains all technology transfers made to Brazilian firms since 1975. The dataset includes the transfer and licensing of patents, trademarks, know-how, and any other industrial knowledge transferred by foreign firms to firms located in Brazil. This dataset is collected from the Brazilian patent office records which, due to legacy regulation, requires firms to register all foreign transfers of industrial knowledge. It includes information on all royalty payments associated with technological upgrade.

The government requires firms to register technology transfer. Before the Central Bank can authorize any international royalty payments, Brazilian firms are obliged to register their technology transfers at the patent office. According to a law dating back to 1962, firms making royalty payments for any intangible capital must register this transfer at the patent office for any royalty payment to a foreign entity to be allowed by the Central Bank. The requirement to register technology transactions at the patent office was created by law no. 4.131 in 1962, during a period of capital controls in Brazil. The goal of the requirement was to limit the payment of royalties and make it more difficult for firms to break the capital control regulation in place at the time. The requirement was maintained after the capital controls were lifted. Consequently, the Brazilian patent office keeps a record of all contracts of technology transfer to Brazilian firms since 1962. The dataset is constructed collecting information from all technology transfers at the patent office.

Coverage. The patent office lists the type of contracts that must be registered before any royalty payment is made. As a general principle, the requirement concerns contracts involving intellectual property transfers that lead to an improvement in the production process or the creation of a new product line. These contracts include the licensing or transfer of patents, trademarks, industrial designs, integrated circuit topography, know-how, or technical services, such as industrial or engineering consulting.

The patent office also clarifies that certain technical services do not classify as technology transfers and are therefore not eligible to be registered. These include financial, marketing, or legal consulting, the licensing or acquisition of software, maintenance services and services that do not generate a technical report.

Firms register technology transfer for legal protection. Because the Brazilian government requires the registration of all technology contracts, firms that do not register their contract lose the possibility of resolving judicial disputes in court. This creates strong incentives on the side of firms to comply with this requirement. According to a survey conducted by [de Souza \(2020\)](#) among experts in intellectual property, companies prefer to register their contracts with the patent office to receive legal protection in case of any dispute.

Variables. For each contract, the dataset contains the name of the foreign firm licensing the technology (“licensor”), its country of origin, the year in which the contract was signed, the name of the Brazilian firm receiving the technology transfer (“licensee”), its tax identifiers, and the type of technology being transferred. For contracts registered after 2010, we also observe a short description of the technology and its monetary value. Using data from the *Relação Anual de Informações Sociais*, an administrative matched employer-employee dataset, we assign a 4-digit sector code from the *Classificação Nacional de Atividades Econômicas 2.0* to each licensee.

Summary statistics and examples. Table 1 shows a breakdown of the contracts by category, as well as information on the number of unique buyers and sellers and the transaction value, when available. Transfer of know-how is the most common type of contract. The following know-how transfers exemplify how Brazilian firms rely on international technology transfers to improve their production processes or introduce new products. In 1983, Gerdau SA, a large Brazilian steel producer, received a technology transfer from Nippon Steel, a Japanese steel producer, to start

Table 1: Summary statistics of technology transfers

Variable	N. Transfers	%
Contract types		
Know-How Transf.	10,928	79.39
Trademark	2,208	16.04
Patent	564	4.10
All	13,765	100
Licensees and licensors		
Unique Licensees	5,484	
Unique Licensors	10,844	
HQ-Branch	401	3.31
Transaction value (in dollars)		
Mean	1,163,047	
Median	645,070	

Notes: The table contains summary statistics of contracts of technology transfer between 1995 and 2015. The top panel contains information on contracts by category, according to the definition used by the patent office. Because technology transfers might fall in more than one category, the percentages do not add up to one. The middle panel contains information on licensees and licensors. The *HQ-Branch* row reports the share of transactions conducted between a headquarters and its branch, identified using information on firm ownership from the National Firm Registry dataset. The bottom panel contains information regarding the value of the transactions.

its production of free cutting steel round bars.⁵ In 2012, *Companhia Brasileira de Vidros Planos*, a Brazilian glass and concrete producer, received technological assistance from *Bottero S.p.A.*, an Italian glass producer, on the design of an assembly line for the production of float glass.⁶

Appendix Figure A.1 shows that, on average, Brazil receives 600 technology transfers per year, with the majority coming from the USA and Germany. Despite the small number, technology transfers are important for large firms. About 18% of the wage bill in 2004 is concentrated among firms that received at least one technology transfer. Studying a 2001 reform that increased the cost for Brazilian firms to receive technology transfers from abroad, [de Souza \(2020\)](#) shows that reduced access to technology transfers has a sizable negative effect on firms' growth.

2.2 Patents and Citations

Information on patents and citations comes from the European Patent Office's Worldwide Patent Statistical Database (PATSTAT, Autumn 2018 version), which collects information from most patent offices around the world, covering both developed and developing countries. For each

⁵This type of steel bar is called "S30VCTS2".

⁶Table 1 also shows that only 3% of technology transfers are made from headquarters to local branches of multinational firms. Therefore, the results that we discuss cannot be explained by firms using different methods of profit repatriation after a tariff change.

patent, PATSTAT records the year of the patent application, the country of the applicants, the technology classes, and the list of cited patents. We link citations to sectors using the crosswalk between International Patent Classification (IPC) fields and sectors created by [Lybbert and Zolas \(2014\)](#).

Summary statistics. Section [A.2](#) shows statistics of Brazilian citations of patents outside of Brazil. Most of the citations are directed to patents in the United States. Chemicals and metallurgy are the most common sectors of the citing patents.

2.3 Firm Ownership and FDI

To measure FDI, we use the Firm Registry, an administrative dataset that contains ownership information for all firms that have ever been active in Brazil since 1990, including currently defunct firms. The dataset reports the list of owners and directors, their country of residence, and the date that they entered ownership of the firm.

Measuring FDI. Using the Firm Registry data we measure FDI as the opening of a new subsidiary, the total acquisition of an exiting firm, or the participation of foreigners in local Brazilian investment. If the foreign firm opens a new subsidiary, we observe the ownership by the parent firm, its country of origin, and the year the company was created. If a foreign firm acquires a Brazilian firm, we observe it entering the ownership of an already existing firm. Finally, if foreign firms or individuals make a partial acquisition of a firm, we also observe a foreign entity entering the ownership of the Brazilian firm.

Summary statistics. Appendix [A.3](#) reports summary statistics on the FDI data. Most FDI is in machinery and chemicals, with the number of firms owned by foreigners growing throughout the sample period. The most common countries of origin of FDI are the United States, Germany, and Italy.

2.4 Other Data Sources

Tariff data come from the World Bank Trade Analysis Information System. Federal procurement data come from the *Controladoria-Geral da União* and cover all the federal procurement since

2000. Data on campaign contributions come from the *Superior Tribunal Eleitoral*, covering the presidential elections of 2002, 2006, and 2010.

3 Empirics

This section describes our identification strategy for the effect of tariffs on cross-country knowledge flows. We estimate a country-by-sector gravity model with a full set of fixed effects. This specification isolates the effect of tariffs on citations coming from exposure to foreign ideas, and the estimated coefficient maps to fundamental parameters in the model. To identify the causal effect, we use the instrumental variable approach proposed by [Boehm et al. \(2023\)](#). As in any difference-in-differences design, this strategy identifies only the relative effect of tariffs on knowledge flows. To quantify the aggregate effects, we develop a model and calibrate it to match the empirical elasticities and aggregate statistics of the Brazilian economy in Sections 5 to 7.

3.1 Empirical Model

Isolating idea diffusion. International trade affects the economy through many channels. It can raise productivity through selection ([Pavcnik, 2002](#), [Melitz, 2003](#)), better inputs ([Fieler et al., 2018](#), [Kugler and Verhoogen, 2009](#)), and access to larger markets ([Atkin et al., 2017](#), [Bustos, 2011](#)). It also affects competition ([Topalova, 2010](#)), investment ([Pierce and Schott, 2018](#)), and innovation incentives ([Bloom et al., 2015](#)). Idea diffusion—the mechanism we study—is only one part of this bundle. A sector-level regression of citations or technology transfers on tariff changes would therefore mix diffusion with these other tariff-induced responses.

We isolate diffusion by estimating a gravity equation in which outcomes are indexed by origin country, sector, and year. This structure lets us compare knowledge flows from different origins within the same narrowly defined industry. Sector–year fixed effects absorb shocks that move all Brazilian firms in a sector at once, including the effect of tariffs on competition, innovation incentives, and others, while origin–year fixed effects absorb changes common to all learning from a given source country in a given year. Identification comes from within-sector shifts in learning across origin countries.

The model provides a direct rationale for this design. In the model, tariffs affect citations and technology transfers through general-equilibrium forces such as competition, selection, and prices, that shift overall activity within a sector but do not determine which origin country’s knowledge

Brazilian firms draw on. By organizing the data at the origin–sector–year level and including sector–year and origin–year fixed effects, the gravity specification nets out these broad responses and isolates the within-sector reallocation of learning across origins. As we formally show in Section 6, this residual variation identifies how tariffs shift learning in specific origin–sector pairs and pins down the model’s diffusion parameters.

Baseline empirical model. We use the following empirical model:

$$y_{n,s,t} = \beta\tau_{n,s,t} + \mu_{n,s}^1 + \mu_{n,t}^2 + \mu_{s,t}^3 + X'_{n,s,t}\kappa + \epsilon_{n,s,t}, \quad (1)$$

where $y_{n,s,t}$ is an outcome, such as citations to country n from firms in sector s or technology transfers from country n to firms in sector s . $\tau_{n,s,t}$ is the average import tariff imposed by the Brazilian government against products in sector s originating from country n in year t . The specification includes a complete set of fixed effects, following the standard in the literature (Head and Mayer, 2014; Boehm et al., 2023). First, we include country–sector fixed effects, $\mu_{n,s}^1$, capturing systematic differences in the level of trade or technology licensing between countries and sectors. Second, we include country–year fixed effects, $\mu_{n,t}^2$, capturing time-varying country-specific shifters. Lastly, we include sector–year fixed effects, $\mu_{s,t}^3$, capturing time-varying sector-specific shifters, such as changes in sectoral demand, market structure, competition or regulation affecting the incentives to adopt technology. The term $X_{n,s,t}$ denotes a set of controls.⁷

Outcomes of interest: citation and technology transfer flows. The main outcomes of interest are the number of technology transfers received by Brazilian firms and citations of foreign patents made by Brazilian firms. Because technology takes time to adjust and has a gradual response to shocks, we aggregate both variables over the following 3 years. Therefore, $y_{n,s,t}$ is the number of citations made by Brazilian firms in sector s to country n over the next 3 years starting at t . Three years is the window commonly used in the literature (Berkes et al., 2022), and the results are robust to alternative windows of aggregation.

Fixed effects control for various sources of bias. The rich set of fixed effects controls for a broad range of sectoral and country-level factors that could bias the estimates of Equation (1).

⁷As controls, we include a three-year lag of the left-hand side variable and a dummy if that market is under MFN tariffs. As robustness checks, we also include tariffs on the inputs and tariffs that other countries impose on Brazilian goods.

The sector–year fixed effect, $\mu_{s,t}^3$, controls for sectoral shocks that might induce tariff changes, such as protection offered to fading industries or lobbying. It also captures the overall effect of tariffs on firms’ incentives to upgrade technology due to increased market access (Bustos, 2011; Atkin et al., 2017), changes in competition (Aghion et al., 2005) or access to superior inputs (Fieler et al., 2018; Kugler and Verhoogen, 2009). Because these forces are common within a sector, their effect is captured by $\mu_{s,t}^3$. The country–year fixed effect, $\mu_{n,t}^2$, captures country-level factors that could lead to more technology transfers. This term should capture rules-of-origin adjustments, standards harmonization, supply-chain deepening, or intellectual property and cooperation provisions. Finally, the country–sector fixed effect $\mu_{n,s}^1$ captures the time-invariant characteristics that influence trade and knowledge flows, such as language and geographical distance.

Instrument: Comparing small MFN partners to preferential partners (Boehm et al., 2023). The estimate of β in Equation (1) may be biased if the government changes tariffs in response to foreign market growth. For example, a foreign breakthrough may lead the government to raise tariffs against that market. Citations may then increase because of both the breakthrough and the tariff change. Because the model cannot separate these effects, the estimate of β could be biased.⁸

To address this endogeneity concern, we use the instrumental-variable strategy proposed by Boehm et al. (2023), which exploits the MFN tariff system under the WTO. WTO rules require Brazil to apply the same MFN tariff to every member country without a preferential trade agreement with Brazil. When the government changes MFN tariffs, possibly in response to shocks affecting large and politically relevant partners, it must change tariffs for all MFN partners, including many small countries that are unlikely to drive policy. This rule creates plausibly exogenous tariff variation for small MFN partners: their tariffs move mechanically with the MFN schedule even though their trade flows and technological growth are too small to affect the government’s decision. We identify the effect of tariffs by comparing outcomes for small MFN partners, whose tariffs fluctuate with MFN changes, with outcomes for partners with preferential trade agreements, whose tariffs in the construction of the instrument are set to a constant. The key identifying assumption is that the government does not choose MFN tariff changes in response to shocks specific to small MFN partners, and we present supporting evidence for this claim.

⁸In Section 6, where we map the model to the empirical results, we show that the key identifying assumption is that changes in tariffs are uncorrelated with changes in the productivity of foreign markets.

Formally, the first stage is given by

$$\tau_{n,s,t} = \tilde{\theta} \mathbb{I}_{n,s,t} \{MFN\} \times \tau_{s,t}^{MFN} + \tilde{\mu} \mathbb{I}_{n,s,t} \{MFN\} + \tilde{\mu}_{n,s}^1 + \tilde{\mu}_{n,t}^2 + \tilde{\mu}_{s,t}^3 + X'_{n,s,t} \tilde{\kappa} + \tilde{\epsilon}_{n,s,t}, \quad (2)$$

where $\mathbb{I}_{n,s,t} \{MFN\}$ is a dummy variable equal to one if products in sector s from country n fall under an MFN trade agreement in year t ,⁹ and $\tau_{s,t}^{MFN}$ represents the MFN tariff applied by Brazil to products in sector s in year t . The fixed effect $\tilde{\mu}$ accounts for the impact of becoming a preferential trade partner. Since becoming a preferential trade partner is an endogenous decision made by the government, we control for this in the second-stage regression as well.

The instrument, $\mathbb{I}_{n,s,t} \{MFN\} \times \tau_{s,t}^{MFN}$, is used in conjunction with the sample selection dropping large trade partners that are under MFN tariffs. Observations are dropped if the following two conditions are satisfied:

$$\mathbb{I}_{n,s} \{MFN \text{ Partner at } t\} \times \mathbb{I}_{n,s} \{MFN \text{ Partner at } t-1\} = 1$$

and

$$\begin{aligned} & \mathbb{I} \{n \text{ is a major partner in } t-1 \text{ in aggregate}\} \\ & + \mathbb{I} \{n \text{ is a major partner in } t-1 \text{ in sector } s\} \\ & + \mathbb{I} \{n \text{ is a major partner in } t \text{ in aggregate}\} \\ & + \mathbb{I} \{n \text{ is a major partner in } t \text{ in sector } s\} > 0, \end{aligned}$$

where, in period t , $\mathbb{I} \{n \text{ is a major partner in } t \text{ in aggregate}\}$ is a dummy equal to one if country n is among the top 10 Brazilian importers and $\mathbb{I} \{n \text{ is a major partner in } t \text{ in sector } s\}$ is a dummy equal to one if country n is among the top 10 Brazilian importers of sector s .¹⁰

Identifying variation and identifying assumption. When using the instrument in Equation (2), the identification of the parameter of interest β in Equation (1) comes from comparing the change in the outcome of interest, such as technology transfers and citations, between small MFN partners and preferential partners after an MFN tariff change. For example, Italy is a small trade partner with Brazil, trading on MFN terms, while Uruguay, with which Brazil has a preferential

⁹Most trade agreements cover all products from a particular country. However, a small subset of agreements applies only to specific products. We consider a sector s to trade under MFN if all its products from country n in sector s are under MFN tariffs. The results are robust to alternative assumptions.

¹⁰In the main part of the paper, we follow the same selection made by [Boehm et al. \(2023\)](#). In the robustness section we show that making more or less strict sample selection does not affect the results statistically or qualitatively.

trade agreement, is exempt from MFN tariffs. To isolate the causal effect of tariffs on technology transfers, we compare the change of technology flows from small MFN partners, like Italy, to those from partners under preferential agreements, such as Uruguay, after Brazil reduced its MFN tariffs, which apply uniformly to all MFN partners.¹¹

The key identifying assumption is that the trends between small MFN partners and preferential agreement partners are parallel. In other words, absent the MFN tariff change, the outcome of interest from both groups would have followed the same trajectory. To validate this assumption, we show that the instrument is unrelated to contemporaneous policies, to origin-market productivity growth, and to innovation activity in the origin markets. We also show that pre-period trends are statistically indistinguishable across the two groups. Taken together, these validation exercises support the parallel-trends assumption.

Impulse response function. To implement the test for parallel trends discussed above and capture the dynamic effect of tariffs on the outcomes of interest, we also estimate the following impulse response function:

$$y_{n,s,t-3+j} = \beta_j \tau_{n,s,t} + \mu_{n,s}^1 + \mu_{n,t}^2 + \mu_{s,t}^3 + X'_{n,s,t} \kappa + \epsilon_{n,s,t}, \quad (3)$$

where $y_{n,s,t-3+j}$ denotes one of our outcomes measured over a three-year window that starts in year $t - 3 + j$: either (i) the stock of patent citations from Brazilian firms in sector s to inventions from country n , or (ii) the stock of international technology transfers from country n to firms in sector s . If the identifying variation is valid, we expect $\beta_j \approx 0$ for all $j < 0$. In estimating Equation (3), $\tau_{n,s,t}$ is instrumented as shown in Equation (2).

3.2 Validation of the Identification Strategy

We validate the instrument with three sets of exercises. First, we show that neither the instrument nor tariffs predict a sector's political connections: they are uncorrelated with federal procurement exposure and campaign contributions, which makes political pressure an unlikely driver of tariff changes for small MFN partners. Second, we show that the instrument does not correlate with the origin market's growth, which is consistent with tariff changes for small MFN partners not

¹¹As it is usual in difference-in-differences, this strategy isolates the relative effect of tariffs on citations and technology transfers. To recover aggregate effects, we calibrate a model to match statistics of the Brazilian economy and the empirical elasticities.

being driven by productivity shocks in those foreign markets. Third, we show that the instruments do not predict innovation or citation activity in the foreign markets, which further suggests that Brazil is not adjusting tariffs to target breakthroughs in small MFN partners.

The instrument does not correlate with other policies in Brazil. Panel A of Table B.1, in the Appendix, shows that the instrument does not predict a sector’s connection to the government. The table reports the regression coefficients of the instrument, averaged at the sector level, on several indicators: the share of firms receiving federal procurement, the share of firms making campaign contributions, total government procurement expenditure, and total campaign contributions by sector. Additionally, Panel A of Appendix Table B.2 shows that tariffs themselves are not correlated with these same measures of government connection, further supporting the absence of political influence on tariff changes.

The instrument does not correlate with the origin market’s productivity. Our identifying assumption is that Brazil does not change tariffs in response to productivity shocks in small MFN partners. Panel B of Appendix Table B.1 evaluates this claim by regressing the instrument on characteristics of the origin market. The instrument does not correlate with sectoral employment, value added, or their respective shares in the origin country’s economy. Panel B of Appendix Table B.2 also shows that tariffs in Brazil themselves do not correlate with the characteristics of the destination country.

The instrument does not correlate with innovation or citations made by other countries. Panel C of Appendix Table B.1 presents a series of placebo tests showing that the instrument and tariffs do not correlated with innovation or citation activity in other countries. Column 1 shows that the instrument does not predict citations from the rest of the world (ROW) to the country-sector pairs affected by Brazilian tariffs. Column 2 shows that the instrument is uncorrelated with patent applications filed in the targeted country-sectors. Columns 3 and 4 extend the analysis to tariff, with a broader sample that includes large trade partners, showing that tariffs remain uncorrelated with both ROW citations and innovation outcomes in the targeted markets. This further supports that Brazil does not systematically change tariffs in response to technological breakthroughs in foreign markets.

4 Empirical Results

In this section, we show that higher import tariffs increase technology transfers to Brazil and that the knowledge embodied in these transfers diffuses through the local economy. Tariffs raise technology transfers, increase patent citations from Brazilian firms to foreign firms, and make Brazilian patents more similar to foreign patents.¹² The citation response is strongest for firms located near domestic technology *licensees* (the recipients of foreign technology), is directed mainly at foreign *licensors* (the providers of that technology), and is followed by substantial firm growth. Moreover, origin country-sectors that increase technology transfers in response to tariffs also experience larger increases in citations from Brazil. These patterns are not explained by FDI, changes in input tariffs, or changes in foreign tariffs against Brazil.

4.1 Tariffs Increase International Knowledge Diffusion

Strong and significant first stage. Appendix Table B.3 reports the estimates of the first-stage regression (Equation 2). There is a strong correlation between the instrument and import tariffs. In all specifications, the F-statistic exceeds 100, well above the threshold for weak instruments.

Tariffs increase technology transfers. Table 2 shows that tariffs increase the number of foreign technologies transferred to Brazil, with the effect driven by new licensors and new licensees entering the market. Column 1 shows the effect of tariffs on the inverse hyperbolic sine of the total number of technology transfers. A 100-percentage-point increase in import tariffs leads to a 12.7% increase in the number of transfers.¹³ The effects on the number of unique licensees (column 2) and licensors (column 3) are similar in magnitude to the estimates in column 1, suggesting that tariffs primarily impact the extensive margin. However, the monetary value of technology transfers responds much more strongly. Column 4 reports results using as outcome the total monetary value of technology transfers.¹⁴ The estimated coefficient exceeds one, indicating that tariff increases disproportionately raise the value of transferred technologies relative to the number of transfer

¹²Because the empirical model is linear, we describe effects as tariff increases for ease of interpretation, even though import tariffs in Brazil fall on average over the sample period.

¹³We follow the standard practice of interpreting changes in the inverse hyperbolic sine as percentage changes, under the common caveats discussed by [Chen and Roth \(2023\)](#). We discuss alternative transformations of the outcome variables in Section 4.5.

¹⁴Technology transfer values are consistently observable only after 2005. To estimate values prior to that year, we approximate them based on three factors: the sector of the recipient firm, the country of the sending firm, and the type of technology transferred. Results are robust to different interpolation methods.

contracts.

Columns 5 to 7 of Table 2 decompose technology transfers by type and show that tariffs increase only know-how transfers, i.e., the transfer of knowledge not subjected to standard forms of intellectual property protection, such as technical consulting, technological assistance, or industrial secrets. Tariffs have no effect on patent licensing (column 5) and a weak negative effect on trademark licensing (column 6), while column 7 shows a strong positive effect on know-how. Transfers of know-how are a common channel of technology diffusion across countries, which previous work has shown to generate large and persistent productivity gains for firms receiving it. For example, [de Souza \(2020\)](#), [Giorcelli \(2019\)](#), and [Giorcelli and Li \(2021\)](#) found that international transfers of know-how had sizable effects on productivity and firm growth in Brazil, Italy, and China, respectively.

Table 2: **Import tariffs and technology transfers**

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>IHS</i>	<i>N.Tech.</i>	<i>N.Unique</i>	<i>N.Unique</i>	<i>Tech.</i>	<i>Patent</i>	<i>Trademark</i>	<i>Know-How</i>
	<i>Transfers</i>	<i>Licensees</i>	<i>Licensors</i>	<i>Value</i>	<i>Licenses</i>	<i>Licenses</i>	
Tariff	0.127*** (0.0472)	0.108*** (0.0409)	0.126*** (0.0469)	1.252*** (0.0450)	0.000475 (0.000974)	-0.0170* (0.0101)	0.143*** (0.0457)
<i>N</i>	1,178,000	1,178,000	1,178,000	1,178,000	1,178,000	1,178,000	1,178,000

Notes: This table reports the coefficients of regressing different measures of technology transfers to Brazil on tariffs, according to the model in Equation (1) using the first-stage regression in Equation (2). All specifications use the inverse hyperbolic sine transformation. The left-hand side variable is the number of technology transfers (column 1), the number of first-time technology licensees (column 2) and licensors (column 3), the monetary value of technology transfers (column 4), the number of technology transfers involving the licensing or reassignment of patents (column 5) and trademarks (column 6), and the number of technology transfers involving the transfer of know-how (column 7). Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

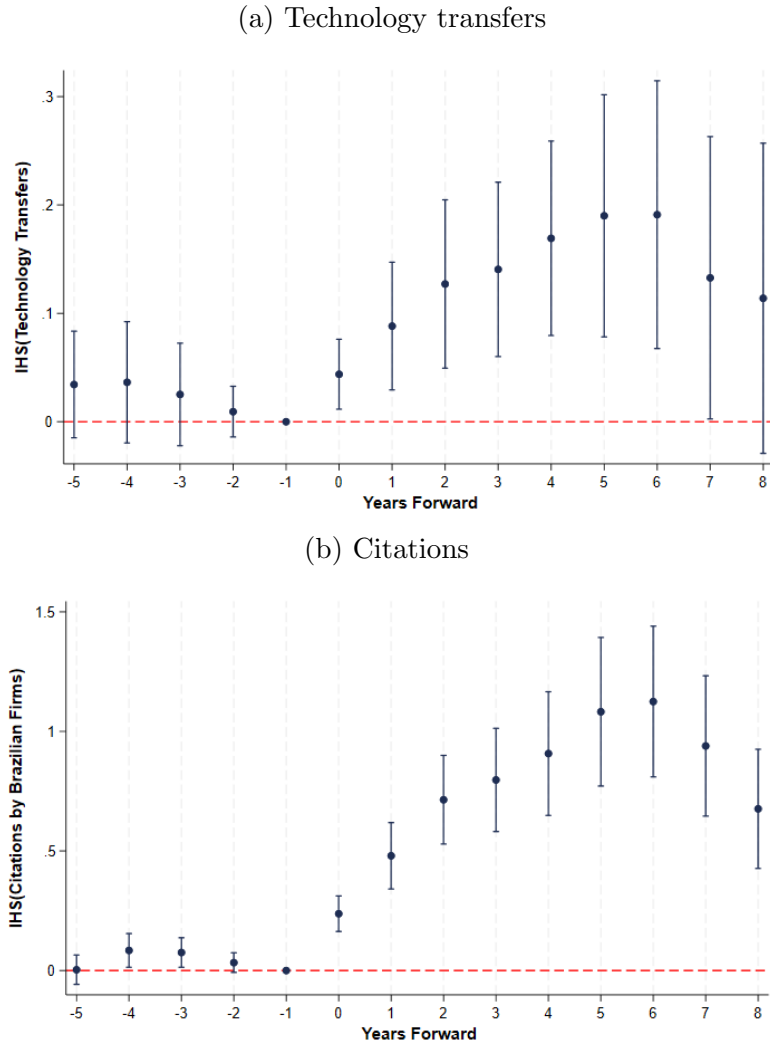
Figure 1a plots the dynamic effects of tariffs on technology transfers. Current tariffs do not predict lagged transfers, which supports our identifying assumption. They do, however, predict higher future transfers: the response builds in the years after the tariff change, but the forward coefficients become statistically insignificant seven to eight years out. This fadeout could come from lower precision and weaker tariff variation at longer horizons as the sample shrinks.

Import tariffs increase citations to foreign patents. Table 3 shows that import tariffs have a positive impact on Brazilian firms' citations to foreign patents. A 100-percentage-point increase in import tariffs raises citations by 71% (column 1) and the probability of observing at least one citation by 53 percentage points (column 2). Column 3 weights citations made to foreign firms by the total citations the Brazilian patent received, a metric of the quality of the local patent, showing

that tariffs increase citations even among influential Brazilian patents. As we show below, these results can be explained by the diffusion among Brazilian firms of foreign knowledge embedded in technology transfers.

Figure 1b shows the effect of tariffs on the dynamics of citations. The effect of tariffs on the path of citations is close to zero in the years before the increase. After the increase, citations rise steadily until year seven and then start declining. The effect appears to be significantly persistent, and, as one would expect, more so than technology transfers since knowledge takes time to diffuse.

Figure 1: **Dynamic effect of tariffs on technology transfers and citations**



Notes: This figure shows the effect of tariffs on technology transfers and citations over different time horizons, as described in Section 3. In the top panel, each dot represents the effect of tariffs on the cumulative sum of technology transfers from $t - 2$ to t . The coefficient at $t = 2$ corresponds to the result shown in column 1 of Table 2. The coefficient at $t = -1$ is omitted because all regressions control for the sum of technology transfers from $t - 3$ to $t - 1$. Similarly, in the bottom panel each dot represents the effect of tariffs on the cumulative sum of citations from $t - 2$ to t , with the coefficient at $t = 2$ matching that in column 1 of Table 3. Again, the coefficient at $t = -1$ is omitted due to the same control for the sum of citations between $t - 3$ and $t - 1$. The bandwidth represents the 90% confidence interval, and standard errors are clustered at the country-sector level.

Import tariffs increase technological similarity to foreign patents. An alternative way of testing for the effect of import tariffs on the diffusion of foreign ideas is by measuring the technological proximity between patents filed by Brazilian firms and foreign patents in each country of origin. Following Bloom et al. (2013), we measure proximity as the cosine similarity between the stock of patents in Brazil and in the foreign country. Formally, technological proximity is defined as

$$Tech. Proximity_{n,s,t} = \frac{W_{s,t} \cdot Y_{n,t}}{\|W_{s,t}\| \|Y_{n,t}\|}, \quad (4)$$

where $W_{s,t} = (W_{1,s,t}, \dots, W_{P,s,t})$ denotes the vector with the share of Brazilian patents in sector s assigned to each technology class $p = 1, \dots, P$ up to year t , and $Y_{n,t} = (Y_{1,n,t}, \dots, Y_{P,n,t})$ denotes the corresponding vector for country n . Equation (4) therefore measures how similar the distribution of patents in sector s is to that of country n : higher values indicate greater similarity.

Table 3: **Tariffs and citations**

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>IHS</i>	<i>At Least</i>	<i>IHS Citations</i>	<i>Technology</i>	<i>IHS. Cit. to</i>	<i>IHS Cit. to</i>
	<i>Citations</i>	<i>One Cit.</i>	<i>Qual.-Weight</i>	<i>Proximity</i>	<i>Licensors</i>	<i>Non-Licensors</i>
Tariff	0.714*** (0.113)	0.537*** (0.0836)	0.627*** (0.192)	0.0570** (0.0277)	0.539*** (0.0886)	0.232*** (0.0769)
<i>N</i>	1,178,000	1,178,000	1,178,000	901,284	1,178,000	1,178,000

Notes: This table reports the coefficients of regressing the number of citations of foreign patents made by Brazilian patents on tariffs using the model in Equation (1) and the first-stage regression in Equation (2). The left-hand side variable is the inverse hyperbolic sine of the number of citations of patents in the foreign country made by Brazilian patents in a given sector (column 1), an indicator equal to one if the sector in Brazil made at least one citation to the foreign country (column 2), the inverse hyperbolic sine of the number of citations of patents in the foreign country made by Brazilian patents in a given sector multiplied by the number of citations received by the Brazilian patent up to three years after its publication (column 3), technology proximity defined in Equation (4), the inverse hyperbolic sine of the number of citations made to technology licensors or patents cited by them (column 5), and the inverse hyperbolic sine of the number of citations to firms that have never made a technology transfer to Brazil or have not been cited by them (column 6). Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Column 4 of Table 3 shows the effect of tariffs on proximity. The point estimate implies that a 100 percentage point increase in import tariffs increases technological proximity by 5.7 percentage points, corresponding to 49% of a standard deviation.

More trade, less diffusion. These results suggest that import tariffs increase the diffusion of ideas across countries. This conclusion is surprising in light of the existing theoretical literature, where prevailing models predict that openness to trade promotes diffusion by facilitating learning from foreign producers and input suppliers. In the next section, we show that the increase in citations to foreign firms is caused by the spread of knowledge embedded in technology transfers,

highlighting a channel through which higher tariffs can lead to greater diffusion of foreign ideas.

4.2 Transferred Technology Diffuses Among Firms

Tariffs increase citations to technology licensors. Columns 5 and 6 of Table 3 separate citations to technology licensors from citations to other firms. The estimate in column 5, which only counts citations to licensors’ patents and to patents they cite, is 2.3 times larger than the effect in column 6, which considers citations of patents that are *not* associated with technology licensors. These results suggest that tariffs lead Brazilian firms to learn and adopt knowledge that is embedded in technology transfers.¹⁵

Table 4: **Tariffs, citations, and characteristics of the citing firm**

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>IHS Cit. by Tech. Licensees</i>	<i>IHS Cit. by Non-Tech. Licensees</i>	<i>IHS Cit. by Importers</i>	<i>IHS Cit. by Non- Importers</i>	<i>IHS Cit. by Exporters</i>	<i>IHS Cit. by Non- Exporters</i>
Tariff	0.183*** (0.0431)	0.582*** (0.107)	0.155*** (0.0443)	0.600*** (0.106)	0.195*** (0.0487)	0.584*** (0.105)
<i>N</i>	1,178,000	1,178,000	1,178,000	1,178,000	1,178,000	1,178,000

Notes: This table reports the coefficients of regressing the number of citations made by Brazilian patents to foreign patents on tariffs using the model in Equation (1) using the first-stage regression 2. The left-hand side variable is the inverse hyperbolic sine of the number of citations made to patents in the foreign country by Brazilian licensees (column 1) and non-licensees (column 2), by importers (column 3) and non-importers (column 4), by exporters (column 5) and non-exporters (column 6). Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

The effect of tariffs on citations is stronger among firms not connected to foreign markets. Table 4 reports the effect of tariffs on citations across different sub-samples of citing Brazilian firms, defined by whether a firm has ever been directly connected to foreign markets. Columns 1 and 2 compare firms that have received a technology transfer with firms that have not; columns 3 and 4 compare importers with non-importers; and columns 5 and 6 compare exporters with non-exporters. Across all three splits, the citation response is about three to four times larger for firms that have never received a technology transfer, imported an input, or exported goods,

¹⁵Notice that, even in the case of transfers taking the form of a patent license, we do not observe the exact patents being licensed, so we cannot track citations of licensed patents directly. Moreover, as we discuss above, the most important type of transfer is know-how, which is industrial knowledge not covered by standard forms of intellectual property protection. Therefore, we interpret citation flows to licensors as measures of the overall diffusion of the knowledge transferred by foreign licensors to Brazilian firms, rather than as exact indicators of the adoption of ideas embedded in specific patents.

Table 5: **Tariffs, citations, and location of the citing firm**

	(1)	(2)
	<i>IHS Cit. Same City</i>	<i>IHS Cit. Diff. City</i>
Tariff	0.443*** (0.0824)	0.243*** (0.0697)
<i>N</i>	1,178,000	1,178,000

Notes: This table reports the coefficients of regressing different measures of citation on tariffs according to the location of the firm citing using the model in Equation (1) using the first-stage regression in Equation (2). In column 1, the left-hand side is the number of citations made by patents of firms on a city with at least one technology licensee during the sample period (excluding technology licensees themselves). In column 2, the left-hand side is the number of citations made by firms in cities that never had a technology licensee. Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

that is, firms without direct exposure to foreign markets.¹⁶

The effect of tariffs on citations is stronger among firms located near technology licensees. The hypothesis that the effect of tariffs on citations is the result of the increase in international technology transfers has a simple testable implication: we should observe larger effects among firms near technology licensees, that is, among firms that are more exposed to the transferred knowledge. To verify this, we separately estimate the effect of tariffs on foreign citations among two groups of firms: those that are geographically close to licensees, and all the remaining firms. Column 1 of Table 5 shows the effect of import tariffs on citations to foreign patents made by firms in the same city of technology licensees (but excluding licensees themselves), while column 2 shows the corresponding effect on citations made by firms in different cities. Consistently with the existence of diffusion of foreign knowledge via technology transfers, we find that the effect among firms located in the same city as a licensee is almost twice as large as the effect among the remaining firms.

Markets providing more technology transfers receive more citations. To link technology transfers and citation flows, we estimate the conditional average treatment effect (CATE) of tariffs on both outcomes. The CATE is the average effect of a tariff change for country-sector pairs that share the same observable characteristics, allowing tariff effects to vary across markets. We then show that the effect of tariffs on technology transfers and on citations are correlated. In other words, markets that respond to tariffs by transferring more technology are also markets that, as an

¹⁶A potential reason why licensees respond less than non-licensees is that, if available, the use of the foreign licensed technology is preferable to in-house innovation, since foreign technologies tend to be more productive (de Souza, 2020).

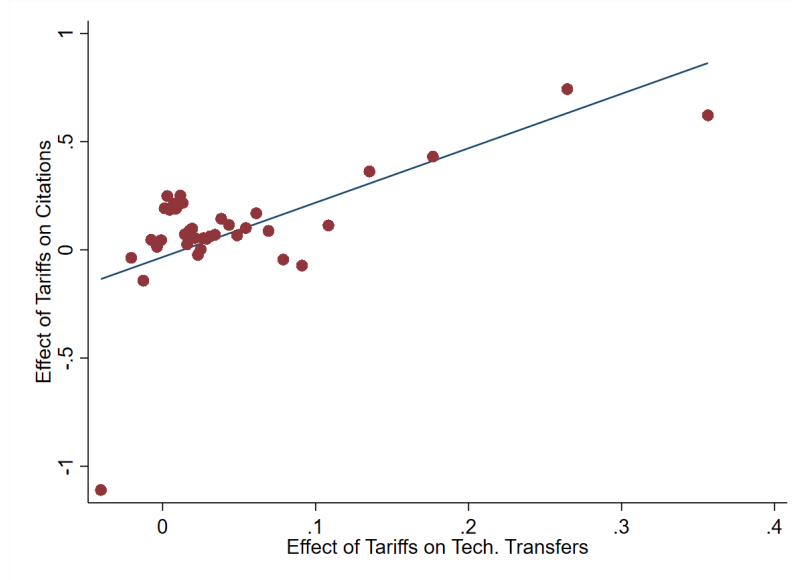
effect of tariffs, receive more citations. This result supports the idea that the diffusion of knowledge embedded in technology transfers is the primary mechanism driving the rise in citations.

We re-write the reduced form version of the empirical model in Equation (1) in long differences:

$$\Delta y_{n,s} = \bar{\beta}(Z_{n,s}) \Delta \tau_{n,s}^{instr} + \tilde{\mu}_n^1 + \tilde{\mu}_s^2 + \epsilon_{n,s}, \quad (5)$$

where $\Delta y_{n,s}$ is the change in outcome $y_{n,s}$ between 1990 and 2010 and $\Delta \tau_{n,s}^{instr}$ is the corresponding change in the instrument.¹⁷ The term $\bar{\beta}(Z_{n,s})$ denotes the CATE of import tariffs on citation or technology transfers to markets with characteristic $Z_{n,s}$. We estimate model (5) using the causal forest method proposed by [Wager and Athey \(2018\)](#). Appendix B.3 describes the procedure in detail.

Figure 2: **Correlation between the effect of tariffs on technology transfers and the effect of tariffs on citations**



Notes: This figure plots the correlation between the conditional average treatment effect (CATE) of the instrument on citations and on technology transfers. The CATE is calculated using causal forest. The treatment effect is conditioned on the stock of patents in the origin country in 1990, the per-capita GDP of the origin country in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from the origin country to sector s until 1990, and the number of citations made by sector s to the origin country n until 1990. Using these variables, we predict the treatment effect for each origin country-sector pair. For clarity, in the figure markets are binned into 40 groups. Appendix B.3 describes the procedure in detail.

Figure 2 plots the CATE of tariffs on technology transfers (horizontal axis) against the CATE of tariffs on citations (vertical axis) for each country-sector pair. For clarity, observations are binned into 40 groups. The figure shows a clear positive relationship: the markets in which tariffs induce

¹⁷We re-write the model in long differences because the standard methods to estimate CATE are not equipped to handle fixed effects. For the same reason, we use the reduced-form model instead of the two-stage least square.

the largest increases in technology transfers are also those in which tariffs generate the largest increases in citations. Conversely, when tariffs do not increase technology transfers, they also have no effect on citations. This pattern indicates that the rise in citations following import-tariff changes is tightly linked to the tariff-induced increase in technology transfers.

Additional results in Appendix B.3 show a large degree of heterogeneity in the effect of tariffs. Tariffs increase citations and technology transfers from high-income countries with a larger stock of patents. These citations are coming from—and technology transfers are going to—Brazilian sectors with a significant history of innovation and technology transfers. The sectors that most benefit from diffusion are the extractive, chemical, and automobile manufacturing industries.

Diffusion through technology transfers. These results indicate that higher tariffs lead foreign firms to transfer their technology to Brazil. Transferred technologies diffuse among Brazilian firms, leading to an increase in citations to technology licensors. This evidence suggests that higher tariffs, by influencing technology transfers, can lead to greater diffusion of technology across countries.

4.3 Knowledge Diffusion is Associated with Firm Growth

To link diffusion to firm growth, we examine how tariff-induced citations correlate with subsequent firm growth. To implement this idea, we decompose the average tariff effect on citations from Table 3 according to the subsequent growth performance of the citing firm. For each origin country n , sector s , time t , we construct an indicator equal to one if at least one firm in sector s both cites a patent from country n and belongs to growth decile d of the within-firm growth distribution over the subsequent three years:

$$\mathbb{I}_{n,s,t}\{\exists i \in s : i \text{ cites } n \text{ and grows in decile } d\} = \beta_d \tau_{n,s,t} + \mu_{n,s}^1 + \mu_{n,t}^2 + \mu_{s,t}^3 + X'_{n,s,t} \kappa + \epsilon_{n,s,t}. \quad (6)$$

We estimate Equation (6) separately for each decile d , using the same baseline specification and instrumental variable strategy as in Equation (1), and plot the resulting coefficients $\hat{\beta}_d$ in Figure 3. A positive and significant $\hat{\beta}_d$ for high deciles means that tariff-induced citations are concentrated among firms that go on to experience exceptionally strong growth—consistent with those firms having genuinely absorbed productive foreign knowledge. Conversely, a near-zero $\hat{\beta}_d$ for low deciles would indicate that tariff-induced citations among slow-growing firms are not statistically different from zero. While these results do not identify the causal effect of diffusion on firm growth, they show how firms that were exposed to foreign ideas due to change in tariffs tend to grow in the

years that follow.

Figure 3 shows that firms learning from foreign ideas due to tariffs experience exceptionally high subsequent growth in both employment and the wage bill. Panel 3a documents that tariffs affect only those citations that are followed by large employment growth: a 100% import tariff raises the probability that a citation is followed by top-decile employment growth by about 20%, but has zero effect outside the top four deciles. Similarly, Panel 3b shows that, following a tariff-induced citation, firms are disproportionately more likely to realize wage-bill growth in the top two deciles of the distribution. These results show that firms more exposed to foreign technology transfers are more likely to have exceptional growth.

Growth from diffusion through technology transfers. This subsection shows evidence that the diffusion caused by tariffs matter for growth. Together, these results indicate that, due to an increase in tariffs, firms receive more technology transfers, which diffuses to nearby firms. After being exposed to foreign technologies, these firms grow faster than in any other moment in their history.

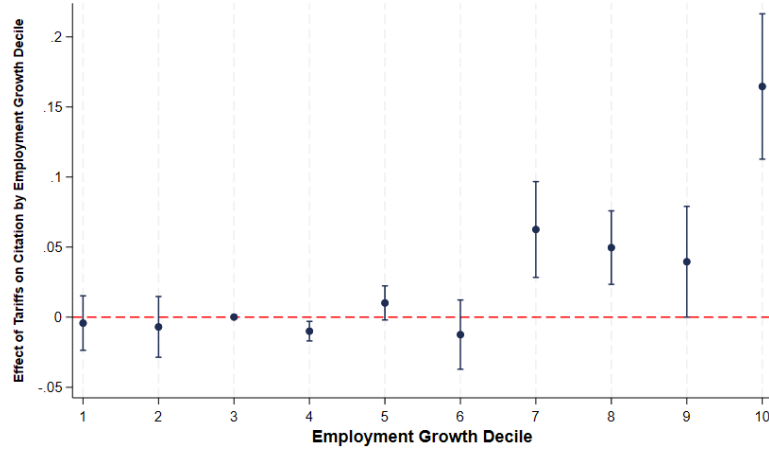
4.4 Alternative Explanations

In this section, we test alternative explanations for the effect of tariffs on technology diffusion. We start by showing that higher tariffs do not increase foreign ownership of Brazilian firms, ruling out FDI as the source of the increase in diffusion. We then show that connection to foreign input suppliers and access to larger foreign markets do not affect technology transfers or citations.

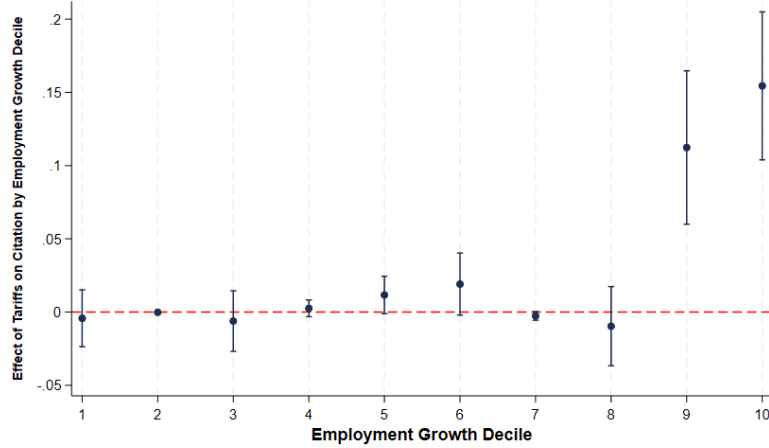
Higher tariffs do not increase FDI. A potential explanation for our findings is that technology transfers are a byproduct of FDI, and higher tariffs promote diffusion by increasing FDI, rather than directly through higher technology transfers. In Panel A of Table 6 we show that higher tariffs do not increase (and, if anything, decrease) measures of foreign ownership of Brazilian firms. Columns 1 to 3 display the effect of tariffs on the number of foreign firm owners, the number of firms owned by at least one foreigner, and an indicator that is equal to one if at least one firm is owned by a foreigner from the country of origin. Columns 4 to 6 display the corresponding estimates when the outcomes are measured in the three years following the tariff change. This result is consistent with a large empirical literature finding null or weak effects of tariffs on FDI (Blonigen, 2005; Blonigen and Piger, 2014; Blonigen, 2002; Wheeler and Mody, 1992; Kinda, 2010; Moder and Spital, 2025;

Figure 3: Citations are Followed by Large Firm Growth

(a) $\mathbb{I}\{\text{Citation by } s \text{ to } n \text{ followed by employment growth in decile } d\}$



(b) $\mathbb{I}\{\text{Citation by } s \text{ to } n \text{ followed by wage bill growth in decile } d\}$



Notes: This figure shows the estimated coefficients from regressing an interaction between a dummy for citation and a dummy for firm-level growth deciles on foreign import tariffs. To construct these variables, we first compute the three-year-ahead growth in employment (Panel A) and the wage bill (Panel B) for each firm. Then, we construct a dummy if in sector s there is a firm citing country n that grow in decile d over the next three years: $\mathbb{I}\{\text{at least one firm in sector } s, \text{ cites } n, \text{ and grow in decile } d\}$. Then, we run a regression on this dummy for each decile d , using specification (1), and the first-stage regression in Equation (2). The x-axis in each figure plots the different deciles while the y-axis plots the estimated coefficients. The bars show the 90% confidence interval. The results illustrate that tariff-induced citations are disproportionately concentrated among firms that subsequently experience exceptional growth.

Matsuura and Hayakawa, 2012). Theory also delivers an ambiguous prediction: higher tariffs can encourage “tariff-jumping” affiliate production, but they can also raise the cost of operating affiliates when production relies on imported intermediates (Ramondo and Rodríguez-Clare, 2013; Helpman, 1985). Our estimates are consistent with these offsetting forces roughly canceling in settings dominated by vertical production links, as emphasized by Blonigen and Piger (2014) and Matsuura and Hayakawa (2012), and with the view that other factors such as institutions and infrastructure may matter more for FDI than tariffs (Kinda, 2010; Matsuura and Hayakawa, 2012; OECD, 2018). Blonigen (2005) reviews this literature and emphasizes that tariff effects are often conflicting across studies, reflecting offsetting mechanisms.

Connection to foreign suppliers does not increase citations. It could be the case that firms learn from foreigners not due to technology transfers but because they import foreign inputs. This may bias our results if tariffs on inputs are correlated with tariffs on outputs. To rule out this potential confounding factor, Panel B of Table 6 (columns 3 and 4) shows the main results, but adds as control the average import tariff on inputs.¹⁸ Tariffs on inputs have no significant effect on technology transfers or on citations. Introducing them as a control in our baseline regression leads to the same conclusions as in our main exercise.

Foreign market access does not affect licensing or diffusion. It could also be the case that the relevant margin is access to foreign markets, as in Bustos (2011), rather than technology licensing. In particular, if import tariffs in Brazil are correlated with the tariffs that other countries impose on Brazilian goods, changes in tariffs can change firms’ incentives to upgrade their technology to better serve foreign markets. To test this explanation, we augment the baseline model by adding as control the import tariff that the country of origin imposes on Brazilian goods in columns 5 and 6 of Panel B in Table 6. This tariff has no effect on technology transfers or citations to foreign patents.

4.5 Robustness

The results presented so far have shown that an increase in import tariffs increases technology transfers and citations to foreign firms. In this sub-section, we show that these results are robust to changing the definition of large trade partners in the construction of the instrument, using

¹⁸Input tariffs are constructed using the Brazilian input-output table from De Souza and Li (2022).

Table 6: Testing for alternative explanations

Panel A: Tariffs and FDI						
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>IHS N.For.</i> <i>Partners</i>	<i>IHS N.Firms</i> <i>F.Owned</i>	$\mathbb{I}(At\ Least$ <i>1 Firm)</i>	<i>IHS N.For.</i> <i>Partners, 3y</i>	<i>IHS N.Firms</i> <i>F.Owned, 3y</i>	$\mathbb{I}(At\ Least$ <i>1 Firm, 3y)</i>
Tariff	-0.00667 (0.0204)	0.00373 (0.0175)	0.00586 (0.0177)	-0.0144 (0.0471)	0.00304 (0.0423)	0.0135 (0.0372)
<i>N</i>	1,010,757	1,010,757	1,010,757	1,010,757	1,010,757	1,010,757
Panel B: Tariffs and Diffusion						
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>IHS N.Tech.</i> <i>Licenses</i>	<i>IHS</i> <i>Citations</i>	<i>IHS N.Tech.</i> <i>Licenses</i>	<i>IHS</i> <i>Citations</i>	<i>IHS N.Tech.</i> <i>Licenses</i>	<i>IHS</i> <i>Citations</i>
Tariff	0.127*** (0.0472)	0.714*** (0.113)	0.205** (0.0802)	0.852*** (0.172)	0.0905** (0.0379)	0.709*** (0.119)
Input tariff			-0.110 (0.0829)	-0.0406 (0.189)		
Tariff on Brazil					-0.000429 (0.000562)	-0.0124* (0.00700)
<i>N</i>	1,178,000	1,178,000	976,976	976,976	599,293	599,293

Notes: Panel A of this table reports the coefficients of regressing different measures of foreign ownership of Brazilian firms on tariffs using the model in Equation (1) using the first-stage regression in Equation (2). The left-hand side is the number of foreign partners in firms of sector s from country n (column 1), the number of firms with at least one foreign partner from country n in sector s (column 2), an indicator equal to one if there is at least one firm with a foreign partner from country n in sector s (column 3), the sum of the number of foreign partners in firms of sector s from country n in the next 3 years (column 4), the number of firms with at least one foreign partner from country n in sector s in the next 3 years (column 5), and an indicator equal to one if there is at least one firm with a foreign partner from country n in sector s in the next 3 years (column 6). Panel B of this table shows the effect of tariffs on technology transfers augmenting the model in Equation (1) with different controls. Columns 1 and 2 display the baseline results. Columns 3 and 4 add as control the average import tariff in Brazil on inputs of each sector, columns 5 and 6 add as control the import tariff imposed against Brazilian products by foreign countries. Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

an Ordinary Least Squares (OLS) regression, varying the set of controls, limiting the sample to high-income countries, and using alternative functional forms to define the outcome variables.

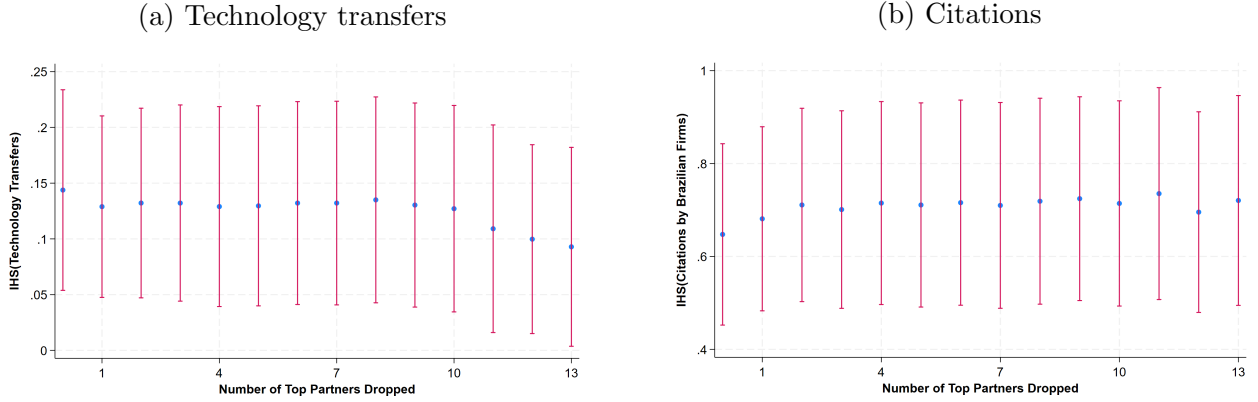
Sample selection of large trade partners. The instrument excludes the top 10 largest trade partners overall and by sector to ensure exogeneity. Figure 4 shows that varying this cutoff—removing between 0 and the top 13 partners—has little impact on the estimated coefficients. The effects of tariffs on technology transfers and citations remain statistically stable across this range, confirming that the results are not sensitive to the specific sample exclusion.

OLS. Appendix Table B.6 presents the main empirical results without using the instrument by Boehm et al. (2023). As such, those regressions include the full sample of countries, including large trade partners. Even without this instrument, we find that tariffs still lead to an increase in technology diffusion and in citations directed toward technology licensors.

Adding and removing controls. Appendix Table B.7 shows the results under different sets of controls. We still find that tariffs increase technology transfers and citations.

Sample selection. Not all countries or sectors are likely to transfer technology or receive citations from Brazilian firms. Appendix Table B.8 shows the results when limiting the sample to high-income countries. As expected, the effect of tariffs on technology transfers and citations are significantly larger in this sub-sample, although with a larger standard error.

Figure 4: **Varying Instrument Selection**



Notes: This figure shows the effect of tariffs on technology transfers (left panel) and citations (right panel) over different sample selection. In the x-axis, we plot the number of top partners dropped from the analysis. For each each estimated coefficient, we drop the top x partners within each sector and across the overall sample. The bandwidth represents the 90% confidence interval. Standard errors are clustered at the country-sector level.

Dealing with zeros. Appendix Table B.9 shows the results using alternative transformations of the outcome variables, following the suggestions by [Chen and Roth \(2023\)](#). Whether we use indicator variables, define the outcome variable as the percentile of technology transfers or citations, or use the raw levels of these variables, we consistently find that higher tariffs increase the diffusion of foreign technologies through technology transfers.

5 Model

In this section, we develop a model of trade, growth, and diffusion that matches our empirical findings and quantifies the effect of tariffs on welfare and growth. The model builds on [Buera and Oberfield \(2020\)](#) by adding two key elements. First, foreign firms choose whether to export to Brazil or to transfer their technology to a Brazilian firm. Second, Brazilian firms learn foreign technologies through two channels: interactions with exporters and interactions with local firms

that received foreign technology transfer. The final growth rate in Brazil depends on the relative strength of these two learning channels.

The model reproduces the empirical patterns and provides a mechanism to interpret how tariffs affect diffusion and growth. Higher tariffs make exporting to Brazil less profitable, so foreign firms shift from exporting toward transferring technology to Brazilian firms. If Brazilian firms learn more from transferred technologies than from interactions with exporters, this reallocation raises the adoption of foreign technology, consistent with the rise in technology transfers and citations documented in the previous section. By boosting learning from foreign firms, tariffs can therefore increase productivity growth. The net effects of tariffs on welfare and growth, however, depend on parameter values, which we calibrate in the next section by matching the empirical elasticities. This calibration also lets us translate the relative effects identified in the data into aggregate effects of tariffs on diffusion in the Brazilian economy.¹⁹

5.1 Environment

The world economy is composed of Brazil (B) and N foreign countries. Time is continuous and indexed by t . Each country n is populated by a mass L^n of workers, each inelastically supplying one unit of labor to active producers within the country. There is a finite number of sectors, indexed by $s \in S$, each composed of a continuum of varieties, indexed by $i \in [0, 1]$. Labor is perfectly mobile across sectors and immobile across countries.

Representative consumer. A representative consumer in each country aggregates individual varieties into sectoral bundles, $C_t^{n,s}$, with a constant elasticity of substitution (CES) ϵ , and sectoral bundles into final consumption, C_t^n , with a constant elasticity of substitution σ :

$$C_t^n = \left[\sum_{s \in S} \left(\int_0^1 c_t^{n,s}(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon(\sigma-1)}{(\epsilon-1)\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \quad (7)$$

The representative consumer within each country values consumption over time according to a logarithmic utility function and a discount rate $r > 0$. Hence, discounted welfare at time 0 is equal to

$$W_0^n = \int_{t=0}^{\infty} e^{-rt} \log(C_t^n) dt. \quad (8)$$

¹⁹Since many elements of the model are standard, in this section we briefly present the setting and equilibrium conditions, and leave the full derivations to Appendix D.

The model has a static and a dynamic component that, for clarity, we present separately.

5.2 Static Equilibrium

Production. Each intermediate variety is produced with labor according to the linear technology

$$y_t^{n,s}(i) = q_t^{n,s}(i)l_t^{n,s}(i), \quad (9)$$

where $q_t^{n,s}(i)$ represents the productivity of the most efficient producer of variety i in country-sector (n, s) (which, in equilibrium, will be the only active producer), and $l_t^{n,s}(i)$ denotes the quantity of labor employed.

Iceberg trade costs and tariffs. Exporting goods from country n to country $m \neq n$ involves a time-varying iceberg trade cost $\kappa_t^{n \rightarrow m, s} \geq 1$, whereby for each unit shipped a fraction $\frac{1}{\kappa_t^{n \rightarrow m, s}}$ arrives at the destination. Notice that trade costs are sector-specific and hence are indexed by s . We further assume, as in [Caliendo and Parro \(2015\)](#), that the trade cost is the product of a time-invariant transportation cost, $\tilde{\kappa}^{n \rightarrow m, s}$, and a time-varying gross tariff, $(1 + \tau_t^{n \rightarrow m, s})$, so that

$$\kappa_t^{n \rightarrow m, s} = \tilde{\kappa}^{n \rightarrow m, s} (1 + \tau_t^{n \rightarrow m, s}), \quad (10)$$

where the tariff payments are collected by the importing country and redistributed as a lump-sum transfer to their representative consumer.

Productivity in Brazil (B). In Brazil (B) there is a large mass of potential entrepreneurs. A subset of them controls a technology to produce variety $i \in [0, 1]$ in sector s . The productivity of the most efficient among them is denoted by $q_t^{B,s}(i)$. As in [Eaton and Kortum \(2002\)](#) and [Buera and Oberfield \(2020\)](#), at time $t = 0$, the distribution of highest-productivity ideas in Brazil is Fréchet with scale parameter $\lambda_t^{B,s} > 0$ and shape parameter $\theta > 1$:

$$Pr \left(q_0^{B,s}(i) \leq q \right) = \exp \left(-\lambda_0^{B,s} q^{-\theta} \right). \quad (11)$$

As we show in [Section 5.3](#), given our assumptions on the diffusion process, this distribution endogenously retains the same Fréchet structure over time, with a time-varying scale parameter $\lambda_t^{B,s}$ and a constant shape parameter θ . To simplify the exposition, we maintain this assumption in the description of the static equilibrium.

Foreign firms choose between exporting or transferring technology to Brazil. Each foreign firm either exports to Brazil or transfers its technology to a Brazilian firm.²⁰ If the producer of variety i in country n and sector s exports, it produces in country n with productivity $q_{X,t}^{n,s}(i)$ and sells at home or ships to Brazil. If instead it transfers its technology, it matches with a Brazilian firm that pays a fee to use the technology and produce in Brazil with productivity $q_{T,t}^{n,s}(i)$. Technology transfers, like trade, face an iceberg cost $\kappa_t^{n \rightarrow B,s}$. For simplicity, we allow technology transfers only to Brazil, not among foreign countries.

Following [Ramondo and Rodríguez-Clare \(2013\)](#) and [Lind and Ramondo \(2023\)](#), we assume that $(q_{T,t}^{n,s}(i), q_{X,t}^{n,s}(i))$ follows a joint Fréchet distribution with common scale $\lambda^{n,s,t}$, shape parameter θ , and correlation governed by ρ :

$$Pr(q_{X,t}^{n,s}(i) \leq q_1, q_{T,t}^{n,s}(i) \leq q_2) = \exp\left(-\lambda_t^{n,s} \left[q_1^{-\frac{\theta}{1-\rho}} + q_2^{-\frac{\theta}{1-\rho}}\right]^{1-\rho}\right), \quad (12)$$

with $\rho \in [0, 1]$. When $\rho = 0$, export productivity and technology-transfer productivity are independent draws; when $\rho = 1$, the productivity of in-house production is proportional to the productivity under technology transfer. Allowing exporting and transferring to draw from distinct (though correlated) productivity distributions captures variation in how well foreign technologies fit the Brazilian context (“appropriateness”), which can differ with environmental conditions ([Moscona and Sastry, 2022](#)) or with the local supply of skills ([de Souza, 2020](#)).

Technology transfer fees. Each potential technology licensor is randomly matched to one potential entrepreneur in Brazil. The licensor then presents a take-it-or-leave-it offer, granting a license to use the technology in exchange for a fixed licensing fee. The offer is accepted as long as the fee does not exceed the profits the entrepreneur can earn from using the technology. As a result, the licensor will set the fee to capture the entirety of the profits generated through the transferred technology.²¹

Two-stage pricing. Firms engage in two-stage pricing (see, e.g., [Acemoglu et al., 2012](#)), which guarantees that the varieties in each country are sold exclusively by the producer with the lowest

²⁰We exclude the possibility of firms doing FDI not because this is an irrelevant mechanism but because we didn’t find evidence of it driving the effect of tariffs on diffusion.

²¹Assuming that the licensor captures the full surplus does not affect the equilibrium allocation, but it does matter for welfare by shifting rents from Brazilian firms to foreign licensors. This is a conservative assumption for our mechanism: it pushes against finding welfare gains from more technology transfers, since it attributes the surplus created by transferred technologies to foreign firms rather than to domestic producers.

marginal cost, charging the optimal markup. In the first stage, each firm has the option to pay a small fee to enter the second stage where price competition occurs. If more than one firm enters the second stage, Bertrand competition ensues, whereby the most productive firm sets the price equal to the marginal cost of the second best producer and captures the full market. Anticipating this outcome, no producer except for the most efficient one enters the second stage. Hence, the best producer sets its price as the optimal markup over its marginal cost.

Domestic production, exporting, and technology transfers. The price faced by the Brazilian representative consumer for variety i in sector s is

$$p_t^{B,s}(i) = \frac{\epsilon}{\epsilon - 1} \times \min \left\{ \frac{w_t^B}{q_t^{B,s}(i)}, \left\{ \frac{w_t^B d_t^{n,s}}{q_{T,t}^{n,s}(i)}, \frac{w_t^n \kappa_t^{n \rightarrow B,s}}{q_{X,t}^{n,s}(i)} \right\}_{n=1,\dots,N} \right\}, \quad (13)$$

where $\frac{\epsilon}{\epsilon-1}$ is the markup and w_t^n is the wage in country n . The minimization chooses the lowest marginal cost of serving the Brazilian market: the first term is production in Brazil with local technology, the second is production in Brazil with technology transferred from country n , and the third is importing the variety from country n .²²

The equilibrium conditions for the foreign countries are analogous to the ones for Brazil, with the exception that we assume that varieties in those countries cannot be provided via technology transfers. As a consequence, the expressions for the equilibrium price index and expenditure shares are identical to the ones in the standard [Eaton and Kortum \(2002\)](#) framework.

5.3 Dynamics of the Productivity Distribution

The dynamic component of the model governs the evolution of the productivity scale parameter, $\lambda_t^{n,s}$. In Brazil, productivity grows as firms learn and adopt better technologies from abroad. The model has two learning channels: firms can learn from foreign technologies embodied in imported goods and from interactions with Brazilian firms that received technology transfers. We allow firms to learn at different rates across these channels.²³ The goal is to capture that firms may

²²The derivation of the equilibrium price index and expenditure shares follows [Lind and Ramondo \(2023\)](#). We leave the details to Appendix [D.1](#).

²³We assume that learning of foreign technologies is the only source of productivity growth in Brazil, hence abstracting from domestic R&D and diffusion of domestic technologies. We make this modeling choice for two reasons. First, this restriction lets us focus on the model's novel mechanism: learning from technology transfers versus learning from imports. Second, the exogenous variation in tariffs identifies learning from foreign sources but not from domestic firms.

learn more efficiently by observing a technology implemented through a transfer than by reverse engineering an imported good. In the next section, we calibrate the learning rate to match our empirical estimates. Outside of Brazil, we assume that $\lambda_t^{n,s}$ grows at a constant and exogenous rate g_λ^* (upper-stars denote variables in their BGP values). We leave all derivations of this section to Appendix D.2.

Contact rate with foreign ideas. There is a continuum measure of potential entrepreneurs in Brazil. Each gets in contact to technologies from foreign exporters of country n and sector s at rate

$$a_{X,t}^{n \rightarrow B,s} = \alpha_t^s \pi_{X,t}^{n \rightarrow B,s},$$

where α_t^s is an exogenous term controlling the overall contact rate of entrepreneurs in sector s at time t . $\pi_{X,t}^{n \rightarrow B,s}$ is the share of sector- s varieties sold in Brazil (B) that originate from exports from country n . Similarly, entrepreneurs might get in contact with licensees of technology transfer at a rate given by:

$$a_{T,t}^{n \rightarrow B,s} = \alpha_t^s \pi_{T,t}^{n \rightarrow B,s},$$

where $\pi_{T,t}^{n \rightarrow B,s}$ is the share of sector- s varieties produced with foreign technology from country n .

Upon contact with a foreign technology from source (n, s) through mode $k \in \{X, T\}$, the entrepreneur draws an idea, q , from the distribution of productivity of active sellers from that source and mode in Brazil.

Personal insight. The idea draw is combined with a personal insight, with the result of this combination giving rise to a new idea. The distribution of the personal insight is such that the Poisson arrival rate of insights larger than z following contact with a foreign idea from source (n, s) and mode k (import or technology transfer) is

$$A_{k,t}^{n \rightarrow B,s}(z) \equiv a_{k,t}^{n \rightarrow B,s} z^{-\theta}, \quad k \in \{X, T\}. \quad (14)$$

Learning from foreigners. The entrepreneur develops a new idea, q' , by combining the personal insight, z , with the foreign idea draw, q . The quality of the new idea, q' , is given by

$$q' = z \zeta_t^{n,s} \omega_k(q)^\beta, \quad (15)$$

where $\zeta_t^{n,s}$ governs the efficiency of learning from source (n, s) at time t , and $\beta \in (0, 1)$ sets the weight on learning in the creation of a new idea. The remaining term, ω_k , captures how learning efficiency varies with the mode through which the entrepreneur is exposed to foreign technology. We normalize $\omega_X = 1$, so ω_T measures the efficiency of learning from technology transfers relative to learning from imports. When $\omega_T > 1$, learning is more efficient through interaction with local users of the technology than through exposure to imported goods. As we discuss further below, ω_T is the key parameter governing the effect of tariffs on learning and on the growth rates.

Law of motion of productivity in Brazil. Under this learning process, the productivity distribution of local producers in Brazil retains its Fréchet structure over time, with a constant shape parameter θ and with a scale parameter, $\lambda_t^{B,s}$, evolving according to the following law of motion:

$$\dot{\lambda}_t^{B,s} = \alpha_t^s \sum_{n=1}^N (\zeta_t^{n,s})^\theta \left[\underbrace{\pi_{X,t}^{n \rightarrow B,s} \mathbf{E}_{X,t}^{n \rightarrow B,s} (q_{X,t}^{n,s})^{\theta\beta}}_{\text{Learning f/ Exporters}} + \underbrace{\omega_T^\theta \pi_{T,t}^{n \rightarrow B,s} \mathbf{E}_{T,t}^{n \rightarrow B,s} (q_{T,t}^{n,s})^{\theta\beta}}_{\text{Learning f/ Tech. Transfers}} \right], \quad (16)$$

where $\mathbf{E}_{X,t}^{n \rightarrow B,s}$ and $\mathbf{E}_{T,t}^{n \rightarrow B,s}$ denote averages over the set of sector- s varieties provided in Brazil via export and technology transfers, respectively, from country n .

Equation (16) illustrates how foreign technology diffusion drives productivity growth in Brazil. The first term inside the brackets captures learning from imported goods, the primary channel emphasized by Buera and Oberfield (2020). This force depends on the share of varieties supplied by exporters, $\pi_{X,t}^{n \rightarrow B,s}$, and their average quality, $\mathbf{E}_{X,t}^{n \rightarrow B,s} (q_{X,t}^{n,s})^{\theta\beta}$. The second term represents learning from Brazilian firms that adopted foreign technology, which we found to be a relevant channel for learning in the empirical section. Growth from this channel depends on the share of varieties using foreign technology, $\pi_{T,t}^{n \rightarrow B,s}$, their average quality, $\mathbf{E}_{T,t}^{n \rightarrow B,s} (q_{T,t}^{n,s})^{\theta\beta}$, and efficiency of learning from transferred technology, ω_T .

The equation also shows how trade policy affects growth through technology transfers. When tariffs increase, the share of varieties imported, $\pi_{X,t}^{n \rightarrow B,s}$, falls, while the share entering through technology transfers, $\pi_{T,t}^{n \rightarrow B,s}$, rises. Brazilian firms then shift from learning from imported goods to learning from local firms that have adopted foreign technologies. The overall impact on productivity growth depends on the relative efficiency of learning from transferred technology, governed by ω_T , and the change in quality of transferred technology, governed by ρ . In the next section, we map this model to the empirical estimates to recover these two parameters.

6 Mapping the Model to the Empirical Results

The quantitative impact of trade liberalization on technology diffusion in the model depends on two key parameters that have no direct counterparts in the existing literature. The first, ρ , governs the degree of substitutability between imports and technology transfers originating from a given country and sector. The second, ω_T , captures the relative efficiency of learning from local users of foreign technologies compared to learning from contact with imported goods. Our reduced-form results suggest that higher tariffs increase technology transfers, implying some substitutability between imports and transfers, and also increase citations to foreign patents, indicating that learning from transferred technologies is relatively more effective.

In this section, we use these empirical patterns to identify ρ and ω_T through a transparent mapping with the reduced-form coefficients. We derive from the model a set of gravity equations that are equivalent to those estimated in the empirical analysis. These equations express the reduced-form relationships between tariffs, technology transfers, and patent citations as functions of the underlying structural parameters, allowing us to recover ρ and ω_T from the empirical coefficients and other observable quantities. In Section 7, using this mapping, we calibrate the full model to evaluate the overall effects of trade liberalization on knowledge diffusion, growth, and welfare.

6.1 Gravity Equation for Technology Transfers: Identifying ρ

We begin by deriving a gravity equation for technology transfers that both rationalizes our empirical specification and provides a way to identify ρ . The key intuition is that higher tariffs raise the cost of serving Brazil through exports, inducing some foreign firms to shift from exporting toward transferring their technology to Brazilian firms. How strongly technology-transfer payments respond to tariffs depends on ρ , which governs the correlation between a firm's export productivity and its technology-transfer productivity. If ρ is higher, meaning that the productivity of exporting and of technology transfer are closer to each other, the elasticity of technology transfers to tariffs increases. This intuition is formalized in the proposition below:

Proposition 1. *Let $V_{T,t}^{n \rightarrow B,s}$ denote the total payments from sector s to country n in exchange for technology transfers at time t . $V_{T,t}^{n \rightarrow B,s}$ can be expressed as*

$$\log(V_{T,t}^{n \rightarrow B,s}) \approx \underbrace{\mu_{n,s}^1 + \mu_{n,t}^2 + \mu_{s,t}^3}_{\text{Fixed effects}} + \underbrace{f_T(\rho|\boldsymbol{\theta}, \mathbf{w}_t)}_{\text{Structural coeff.}} \tau_t^{n \rightarrow B,s} + \underbrace{\varepsilon_{T,t}^{n,s}}_{\substack{\text{error term} \\ (\text{zero mean})}}, \quad (17)$$

where $\boldsymbol{\theta}$ is a vector of common time-invariant parameters, $\mathbf{w}_{\bar{t}}$ is a vector of equilibrium quantities at the baseline. The function $f_T(\cdot)$ can be expressed in closed-form.

This proposition provides the key mapping that allows us to identify ρ . Conditional on the full set of fixed effects, the effect of tariffs on the value of technology transfers can be written as a closed-form function of ρ , known structural parameters, and observables such as equilibrium wages. After calibrating these parameters to standard values in the literature, we use the elasticity in Section 4 to pin down ρ .

Identifying assumption: $\tau_t^{n \rightarrow B, s} \perp \textit{foreign productivity shocks}$ Proposition 1 also makes the identifying assumption explicit. To identify the structural coefficient, tariffs must be orthogonal to the error term. As shown in Appendix D.3.1, the error term, $\varepsilon_{T,t}^{n,s}$, includes unobserved shocks to foreign productivity. Thus, if tariffs or the instrument are uncorrelated with growth in other markets—as we document in Section 3.2—the tariff effect on technology transfers identifies ρ .

Gravity structure isolates ρ from general equilibrium effects. Finally, Proposition 1 highlights the importance of the gravity structure. Aggregate and sector-level variables that respond to tariff changes, such as wages and sectoral prices, are absorbed by the fixed effects, so identification relies on within-fixed-effects variation.

6.2 Gravity Equation for Patent Citations: Identifying ω_T

Mapping citations in the model to the data. Similarly to Kortum (1997), we assume that a new idea is patented when its quality q' exceeds the sector- and time-specific threshold δ_t^s . A patent counts as a citation to country n by sector s when it builds on technology from (n, s) . Let $\text{cit}_t^{n,s}$ denote total citations from sector s to country n at time t .

Citations are informative about ω_T . The effect of tariffs on citations is informative about ω_T , the learning efficiency from transferred technology. When tariffs increase, technology transfers rise, shifting firms' learning away from imports and toward transferred technologies. If citations rise with tariffs, it implies that firms learn more from technology transfers than from imports, so $\omega_T > 1$. If citations fall with tariffs, the opposite holds, implying $\omega_T < 1$. The proposition below formalizes this intuition.²⁴

²⁴Proof is provided in Appendix D.3.2.

Proposition 2. *The quantity $\text{cit}_t^{n,s}$ can be expressed as*

$$\log(\text{cit}_t^{n,s}) \approx \underbrace{\mu_{n,s}^1 + \mu_{n,t}^2 + \mu_{s,t}^3}_{\text{Fixed effects}} + \underbrace{f_{\text{cit}}(\omega_T | \rho, \boldsymbol{\theta}, \mathbf{w}_t)}_{\text{Structural coeff.}} \tau_t^{n \rightarrow B, s} + \underbrace{\varepsilon_{\text{cit},t}^{n,s}}_{\substack{\text{error term} \\ \text{(zero mean)}}}, \quad (18)$$

where $\boldsymbol{\theta}$ is a vector of common time-invariant parameters, and $\mathbf{w}_t$ is a vector of equilibrium quantities at the baseline. The function $f_{\text{cit}}(\cdot)$ can be expressed in closed-form.

Proposition 2 maps the empirical tariff elasticity of citations to ω_T . Because $f_{\text{cit}}(\cdot)$ is closed form and invertible, we can solve for ω_T by matching the model-implied coefficient in Equation (18) to its empirical counterpart.

Identifying assumption and importance of gravity structure. In Appendix D.3.2, we show that the error term, $\varepsilon_{\text{cit},t}^{n,s}$, captures shocks to the productivity of source (n, s) and to the efficiency of learning from that source. Identification therefore requires tariff changes to be orthogonal to these unobserved determinants of knowledge diffusion. In support of this identifying assumption, in the empirical analysis we showed that the instrument does not correlate with citations from other countries to market (n, s) .

7 Quantitative Exploration

In this section, we discipline the key learning mechanism in the model using the empirical responses of technology transfers and patent citations to tariff changes, together with the gravity-based mapping between model and empirical estimates developed in Section 6. We then use the calibrated framework to quantify the implications of the wave of tariff reductions implemented in Brazil in the early 1990s for productivity growth and welfare, and to study the implications of our mechanism for policies targeting technology transfers.

7.1 Calibration

For the quantitative exploration, we work with a discrete-time version of the model calibrated at an annual frequency. To simplify the environment, we group foreign countries into two sets: high-income (H) and low-income (D). Since we focus exclusively on domestic outcomes in Brazil, we assume that trade is frictionless within each set, and therefore refer to them simply as the

high-income and low-income foreign country, respectively. Because technology transfers from low-income countries are negligible in the data, we further assume that all technology transfers originate from the high-income foreign country (i.e., we set $d_t^{D,s} = \infty$ for all s and t).

Our calibration proceeds in three steps. First, we calibrate a subset of parameters to common values from the literature. Second, using data on sectoral value added, trade flows, and technology transfers, we calibrate the BGP of the model in 1990, before the onset of the wave of trade liberalization. Third, we pin down the values of the two key parameters, ρ and ω_T , using the gravity-based approach described in Section 6.

Step 1: Externally assigned parameters. We assign standard values from the literature to a subset of the parameters. Following Buera and Oberfield (2020), we set θ , the shape parameter, to 4 and β , which governs the strength of diffusion, to 0.6. The elasticity of substitution across sectors, σ , is set to 2, which is in the ballpark of values used in quantitative macro and trade models featuring nested CES structures (e.g., Atkeson and Burstein, 2008; De Souza and Li, 2022). The elasticity of substitution of across varieties, ϵ , is set to 2.94, implying a labor share, $\frac{\epsilon-1}{\epsilon}$, of 66%. The yearly discount rate r is set to 5%. The top panel of Table 7 summarizes this assignment of parameters.

Step 2: Initial balanced growth path. We initialize the model by assuming that the world economy is in a BGP in which productivity grows at the same rate in all countries and sectors and, as a result, relative prices, sectoral employment shares, and expenditure shares are constant. We set g_λ^* (the growth rate of $\lambda_t^{n,s}$ in foreign countries) so that, along the BGP, the growth rate of income per capita is 2% per year. Given this, in order to have a BGP, the composite term α_t^s must grow at a rate $g_\alpha = (1 - \beta)g_\lambda^*$.²⁵ We assume that trade costs are symmetric in the initial BGP (i.e., $\kappa_t^{n \rightarrow m,s} = \kappa_t^{m \rightarrow n,s}$). Given a guess for ρ (which will be determined below), we calibrate the initial BGP by setting the initial values of $\lambda_t^{n,s}$, $\kappa_t^{n \rightarrow m,s}$, and $d_t^{H,s}$ to exactly match the 1990 relative average income by country, value-added and import shares by sector and country, as well as the share of labor in each sector employed by Brazilian firms receiving technology transfers from foreign firms. We set tariffs for each sector to their respective averages in 1990.

²⁵Notice that, given initial values for $\lambda_0^{H,s}$ and $\lambda_0^{B,s}$, we can solve for all the stationary objects and then back-out the value of α_0^s that is consistent with balanced growth. In other words, we can always initialize the model to be on a BGP by choice of α_0^s .

Step 3: Calibrating ρ and ω_T . Within the initial BGP, we use the gravity-based approach described in Section 6 to infer the two time-invariant parameters that govern the relevance of technology transfers: ρ and ω_T . Propositions 1 and 2 provide closed-form expressions for the reduced-form coefficients of technology transfers and patent citations with respect to tariffs. Given the externally set parameters and equilibrium quantities, these expressions only depend on ρ and ω_T and therefore allow us to identify both parameters.

We begin by setting ρ so that the model-implied expression $f_T(\rho|\boldsymbol{\theta}, \mathbf{w}^*)$, where $\mathbf{w}^*$ denotes the set of equilibrium quantities in the initial BGP, matches the reduced-form coefficient of the value of technology transfers on tariffs (column 4 of Table 2), which is estimated to be 1.25.²⁶ Intuitively, a larger value of ρ implies a higher correlation between the productivity distributions for exporting and for technology transfers. As a result, a small increase in trade costs induces a sizable reallocation of activity toward technology transfers. The calibration yields a value of $\rho = 0.637$. This suggests that productivity draws are moderately, but not perfectly, correlated across exporting and technology transfers: the most advanced technologies abroad are not necessarily those that are most productive in Brazil.

Next, we set ω_T so that the model-implied expression $f_{\text{cit}}(\omega_T|\rho, \boldsymbol{\theta}, \mathbf{w}^*)$ matches the reduced-form coefficient of the number of citations on tariffs (column 1 of Table 3), which is estimated to be 0.71. Intuitively, a higher value of ω_T implies that learning from technology transfers is more effective relative to learning from imported goods. As a result, when tariffs rise and foreign firms reallocate activity from exporting toward providing technology transfers, the overall learning gains from foreign knowledge increase, leading to a larger rise in citations. The calibration yields $\omega_T = 1.43$, which points to a substantial learning advantage associated with technology transfers relative to exposure through imports.

Appendix Figure C.16 illustrates the identification by plotting $f_T(\cdot)$ for different values of ρ (left panel) and $f_{\text{cit}}(\cdot)$ for different values of ω_T (right panel). The plots show that the calibration strategy identifies unique values of the parameters that are consistent with the empirical moments. The values and moment fit are summarized in the bottom panel of Table 7.

²⁶See Appendix D.3.3 for details on how we compute $f_T(\cdot)$ and $f_{\text{cit}}(\cdot)$ for the sample and sectoral disaggregation used in the quantitative analysis.

Table 7: **Parameter values and targets**

<i>Parameter</i>	<i>Description</i>	<i>Value</i>	<i>Target</i>
<i>Externally assigned parameters</i>			
θ	Fréchet shape parameter	4.0	Buera and Oberfield (2020)
β	Learning parameter	0.6	Buera and Oberfield (2020)
σ	Elasticity of subs. across sectors	2	Within common range
ϵ	Elasticity of subs. across varieties	2.94	Labor share 66%
g_λ^*	Growth rate of the technology frontier	0.0824	Yearly income growth 2%
r	Discount rate	0.05	Within common range
<i>Internally calibrated parameters</i>			
			Model Data
ρ	Correl. between exp. and tech. transf.	0.637	Table 2, column 4 1.25 1.25
ω_T	Learning efficiency tech. transf.	1.43	Table 3, column 1 0.71 0.71

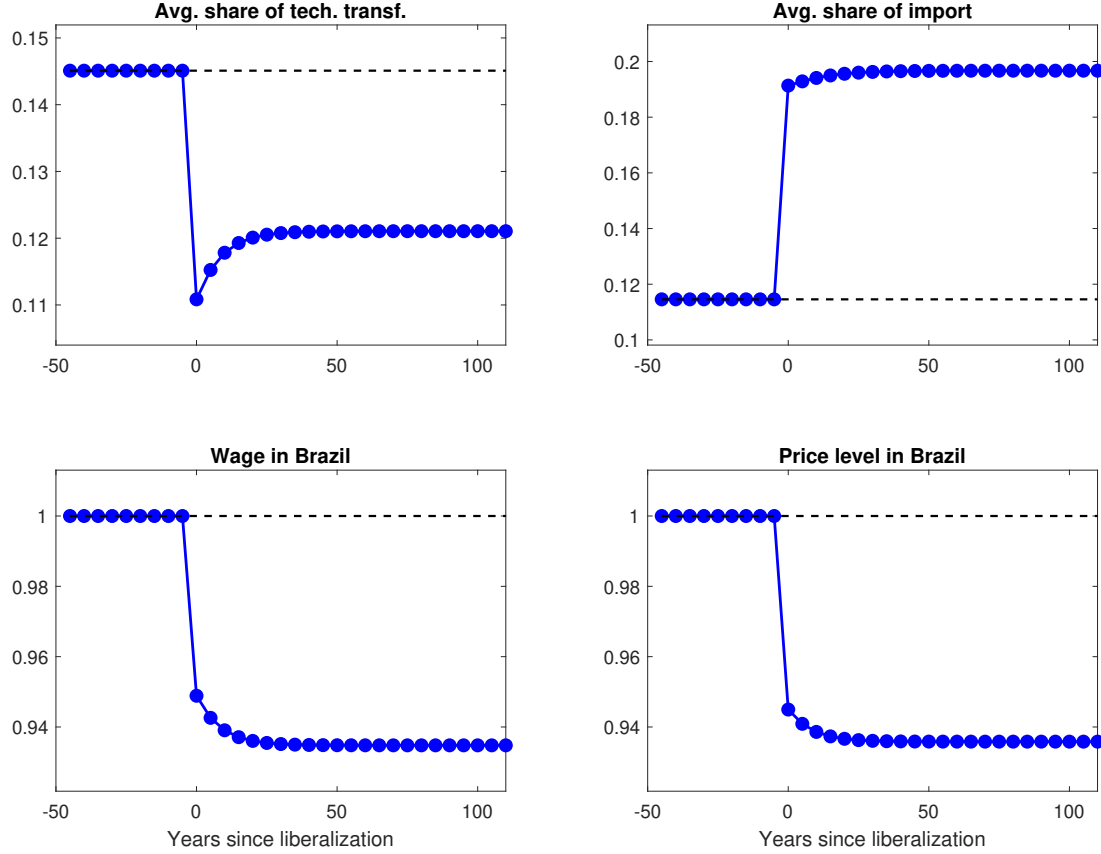
7.2 Quantitative Results

We now quantify how trade liberalization affects welfare once we account for technology transfers and their role in diffusion. We implement in the model the same change in tariffs that we observed in the data and compare the previous BGP to the actual equilibrium. Trade liberalization raises welfare on impact by reducing trade distortions, but it comes with a long-run cost: higher openness weakens foreign firms' incentives to transfer technology, and because Brazilian firms learn more from technology transfers than from imports ($\omega_T > 1$), this shift slows diffusion and makes long-run gains smaller than short-run gains. The magnitude of this mechanism is large: the overall welfare gains from liberalization are 0.27% in the model with technology transfers, but 5.98% in a counterfactual without technology transfers, implying that ignoring technology transfer substantially overstates the dynamic gains from trade. Finally, because technology transfers generate diffusion spillovers that private agents do not internalize, the optimal policy combines liberalization with a subsidy to technology transfers, which restores transfers and accelerates diffusion.

7.2.1 Dynamic effects of trade liberalization

Liberalization decreased technology transfer share but increased import share. Figure 5 shows the effect of the trade liberalization on technology transfers, imports, wages, and prices. When tariffs fall, exporting to Brazil becomes more profitable than transferring technology, so foreign firms shift toward exports. As a result, the import share rises and the expenditure share

Figure 5: Dynamic effects of the liberalization



Notes: The marked blue lines show the expenditure share of technology transfers (top-left panel), the expenditure share of imports (top-right panel), the wage rate (bottom-left panel), and the price level (bottom-right panel) in Brazil.

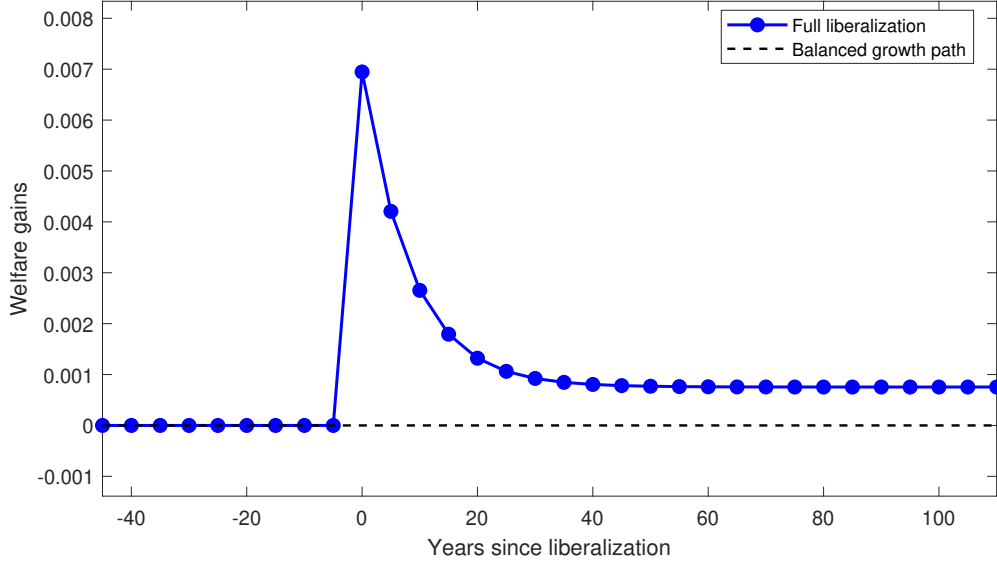
on varieties produced through technology transfer falls.²⁷ Lower tariffs and the substitution of local varieties by imported ones reduce both wages and the price level in Brazil.

Liberalization: static gains but dynamic losses. Figure 6 summarizes the impact of the trade liberalization on the welfare of the representative consumer. The dotted blue line shows the percentage deviation of real consumption under the full liberalization from a baseline scenario where the economy remains on its BGP without liberalization.

On impact, tariff reductions generate static gains from trade: lower import prices reduce the cost of the consumption basket and raise real wages and consumption. In our experiment, consumption rises by 0.69% relative to the initial BGP, in line with standard trade models (Arkolakis et al., 2012). In the years immediately following the trade liberalization, the diffusion response has

²⁷The response of technology transfers is more pronounced in the short run than in the long run: as the gap between Brazil's accumulated knowledge and the foreign technology frontier widens over time, the returns to adopting foreign technologies increase, partially offsetting the initial drop in technology transfers.

Figure 6: **Welfare gains from liberalization**



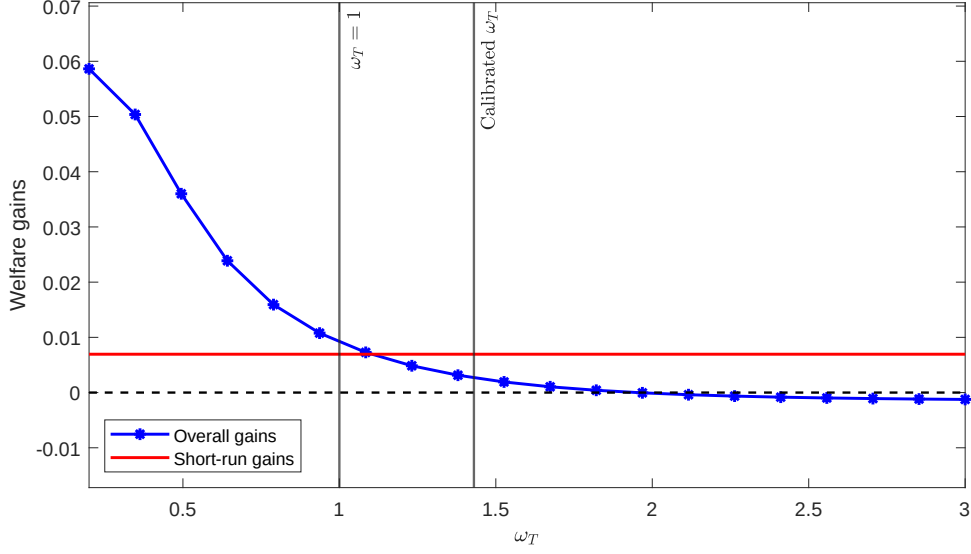
Notes: The marked blue line shows consumption under the full liberalization scenario in percentage deviation from the BGP without liberalization.

not yet materialized, so the welfare response is dominated by these static gains of reducing trade distortions.

Over time, however, the liberalization weakens foreign firms' incentives to transfer technology, slowing the diffusion of foreign ideas that drives productivity growth. As a result, welfare gains shrink after the initial jump, unlike a large class of trade-and-growth models that do not have the technology transfer margin (e.g., [Sampson, 2016](#), [Buera and Oberfield, 2020](#), [Perla et al., 2021](#)). Welfare remains above the no-liberalization baseline throughout the transition and in the new BGP, but the full dynamic gains are equivalent to a permanent 0.27% increase in consumption per period. The transition is relatively fast, with a half-life of about 6.5 years, comparable to the neoclassical growth model ([King and Rebelo, 1993](#)).

Efficiency of learning, identified from micro-data, drives the long-run effects. The parameter ω_T , which controls learning from technology transfers relative to learning from imported goods, is a key determinant of the dynamic gains from trade. Figure 7 illustrates this role by plotting the overall welfare gains from liberalization (blue line) for a range of values of ω_T . When ω_T is high, firms learn more efficiently from technology transfers than from imports. Because liberalization reduces technology transfers, a higher ω_T makes the dynamic welfare cost of liberalization larger, as the economy forgoes a more productive learning channel. When ω_T is low, firms learn more from imported goods than from technology transfers, so liberalization delivers larger welfare

Figure 7: **Welfare gains from liberalization for different degrees of diffusion via transfers**



Notes: Short-run (solid red line) and overall (marked blue line) welfare gains, in percentage deviation from the BGP, from the liberalization for a range of values of ω_T .

gains.

Figure 7 also plots the calibrated value of ω_T . In the model, the citation elasticity identifies the relative learning efficiency of technology transfers versus imports. Because we found tariffs to increase citations in the data, the model implies that learning from technology transfers is more efficient than learning from imports, so $\omega_T > 1$. We recover ω_T by matching the model-implied citation elasticity to its empirical counterpart.

The importance of idea diffusion. Table 8 highlights the role of idea diffusion in shaping the long-run outcomes after liberalization. The table reports average productivity (both unconditional and conditional on active producers in Brazil) and real consumption in the new BGP following tariff reductions relative to the initial BGP. Column 1 presents results from the full model, while column 2 shows a version without the diffusion channel. In the full model, liberalization reduces average productivity due to lower learning by 0.31%. Conditional productivity rises in both models due to selection, although the loss in diffusion makes the rise negligible in the full model. As a result, the rise in real consumption is significantly lower in the full model compared to the model without diffusion (0.08% vs. 0.69%), where the welfare gain aligns with the estimates of Costinot and Rodríguez-Clare (2014).

Table 8: **Importance of idea diffusion in the Brazilian trade liberalization**

	(1)	(2)
	Full model	No diffusion
Average productivity:		
$\Delta \log \mathbf{E} [q^B]$	-0.31%	0
Productivity active domestic firms:		
$\Delta \log \mathbf{E}^{B \rightarrow B} [q^B]$	~ 0	+0.28%
Real Consumption in Brazil:		
$\Delta \log \mathcal{C}_t^B$	+0.08%	+0.69%

Notes: The table reports long-run outcomes (i.e., in the new BGP) in Brazil following the trade liberalization. $\mathbf{E} [q^B]$ denotes the average productivity of local producers, where each sector is weighted by the share of labor before the liberalization. $\mathbf{E}^{B \rightarrow B} [q^B]$ denotes the corresponding average across active producers only. Column 1 reports outcomes in the full model, where diffusion occurs through contact with exporters and with licensees. Column 2 reports outcomes in the model where productivity growth g_λ^B is exogenous.

7.2.2 Optimal subsidy to technology transfers.

This diffusion process implies that technology transfers generate a dynamic effect on local productivity growth that is not internalized by the providers and receivers of the technology. This dynamic externality opens up room for policy intervention. In this section, we show that a technology transfer subsidy of 15.4% together with the trade liberalization would maximize welfare, leading to an increase in consumption by 1.42%.

We introduce subsidies to technology transfers in the form of multiplicative discounts on the iceberg cost of transfers, financed via a lump-sum tax on the representative consumer in Brazil. Denoting by $\bar{d}_t^s$ the base iceberg cost in the absence of subsidies, the net iceberg cost d_t^s is defined as

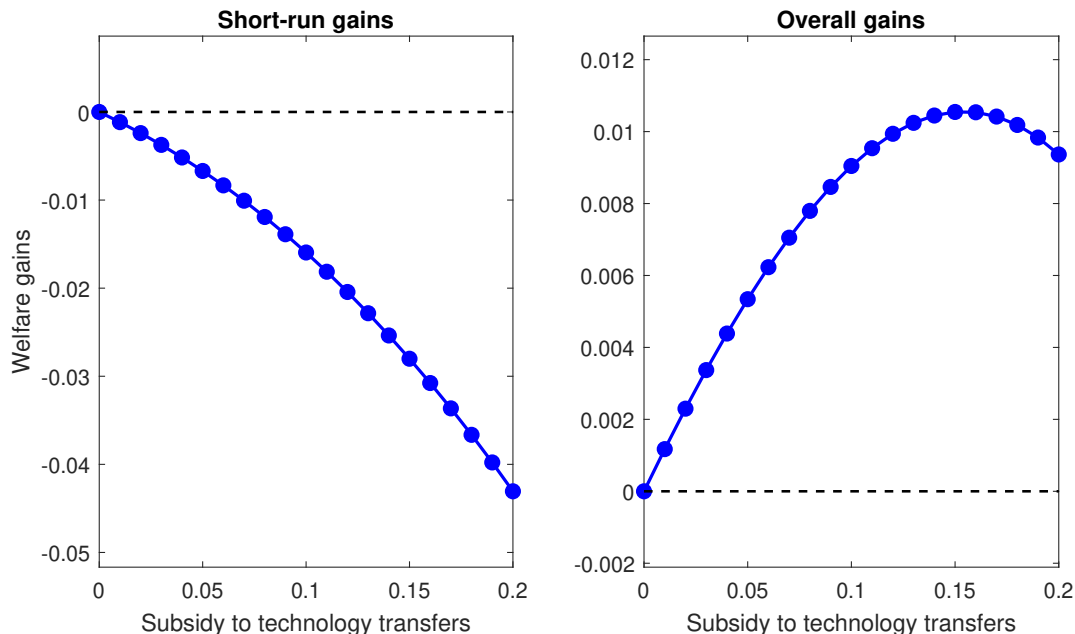
$$d_t^s = \bar{d}_t^s (1 - \bar{S}_t),$$

where $\bar{S}_t < 1$ denotes the subsidy rate. We assume that the subsidy introduction is permanent and is unexpectedly introduced after a given period $\bar{t}$.²⁸

Other things being equal, a positive subsidy makes technology transfers more profitable, promoting a reallocation of local labor and local demand towards varieties produced locally using foreign technology. In the short run, this reallocation may generate static distortions that reduce social welfare. However, this negative impact may be compensated for by the effect of transfers

²⁸In other words, there is a period $\bar{t}$ such that $\bar{S}_t = 0$ for $t < \bar{t}$, and $\bar{S}_t = \bar{S} \geq 0$ for $t \geq \bar{t}$. We have numerically verified that optimizing the dynamic path of subsidies and their distribution across sectors does not generate substantial changes in welfare compared to a uniform and permanent subsidy. Therefore, we focus on this simpler subsidy scheme.

Figure 8: **Short-run and overall welfare gains from subsidizing technology transfers**



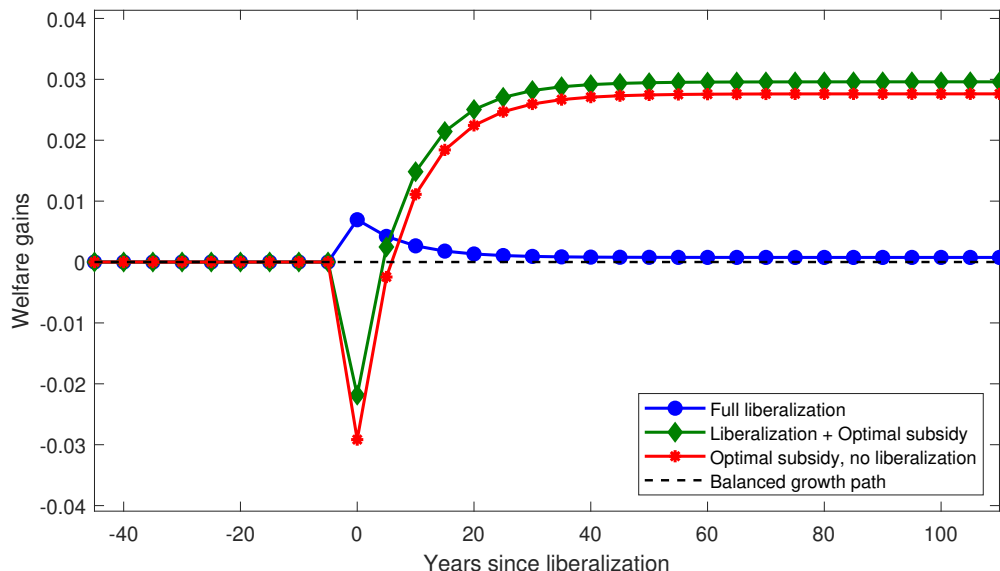
Notes: The left panel shows instantaneous consumption in the period in which the subsidy is introduced, in percentage deviation from the BGP. The right panel shows total discounted welfare in consumption-equivalent percentage deviation from the BGP.

on local learning and knowledge accumulation, generating positive dynamic welfare gains. The balance between static losses and dynamic gains defines an optimal level of subsidies.

Figure 8 illustrates this point. The left panel shows the short-run welfare effect (that is, the percentage change in consumption in the period in which the policy is introduced) induced by the range of subsidy rates on the horizontal axis. Subsidies to transfers generate substantial static losses that become progressively larger as the subsidy rate increases. The right panel depicts the overall gains, which account for the static distortion as well as the positive dynamic effect on learning and productivity growth. The overall gains display an inverted-u shape: Higher subsidy rates increase total discounted welfare up to an optimal subsidy rate, after which the static distortion dominates, and increasing the subsidy rate reduces total welfare. This combination of forces gives rise to an optimal subsidy rate, which we find to be equal to 15.4%. The corresponding welfare gains are equal to 1.06% of consumption per period.

Figure 9 displays the full dynamics of welfare when subsidies of technology transfers are introduced. The starred red line shows the dynamics with the optimal subsidy rate but no concurrent liberalization. Contrary to those induced by the liberalization (which are depicted in the marked blue line), these dynamics show a sharp drop in welfare on impact (-2.92%), followed by higher growth and large long-run gains that more than compensate for the short-run losses. In the long

Figure 9: **Welfare gains from optimal subsidy to technology transfer**



Notes: The figure shows instantaneous consumption in each period under the full liberalization scenario (marked blue line), under the optimal subsidy to transfers and no liberalization (starred red line), and under the combined liberalization and subsidy scenario (diamond-marked green line), in percentage deviation from the BGP.

run, consumption is 2.76% higher compared to the initial BGP. The diamond-marked green line in the same graph shows the combined effect of the optimal subsidy and the liberalization being implemented at the same time. The effect of the two policies is nearly additive. Despite a 2.19% drop in welfare in the short run, the combined policies generate large long-run gains, with an overall effect on total discounted welfare of 1.42%, improving the overall gain from liberalization by a factor of five.

8 Conclusions

Using a unique dataset on technology transfer contracts between foreign and Brazilian firms, we examined how trade policy affects technology diffusion. Our empirical findings show that higher tariffs significantly increase technology transfers and knowledge spillovers, as measured by patent citations, particularly among firms with limited foreign exposure and those near technology transfer recipients. The rise in citations is concentrated among foreign firms that previously transferred technology to Brazil, suggesting that these transfers play a crucial role in facilitating knowledge diffusion within the domestic economy. Importantly, we rule out alternative explanations, such as foreign direct investment or supplier relationships, as the main drivers of this effect, reinforcing the conclusion that technology transfers are a key channel through which trade policy shapes

knowledge diffusion.

These findings challenge the conventional view that trade liberalization always improves access to foreign ideas, showing instead that lower tariffs can reduce opportunities for domestic firms to absorb foreign knowledge. To explore the welfare implications of this result, we extended a trade and idea diffusion model to explicitly incorporate technology transfers and their role in knowledge diffusion. We calibrated the model to match the empirical responses of technology transfers and patent citations to changes in tariffs. Our quantitative results suggest that accounting for technology transfers significantly reduces the long-run welfare gains from trade liberalization, with long-term gains from openness shown to be lower than short-term gains. However, this result also highlights an opportunity for policy intervention due to the existence of a dynamic externality in technology transfers: combining trade liberalization with targeted subsidies for technology transfers can substantially enhance long-run welfare.

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Online Appendix
(for online publication only)

More Trade, Less Diffusion: Technology Transfers and the Dynamic
Effects of Import Liberalization

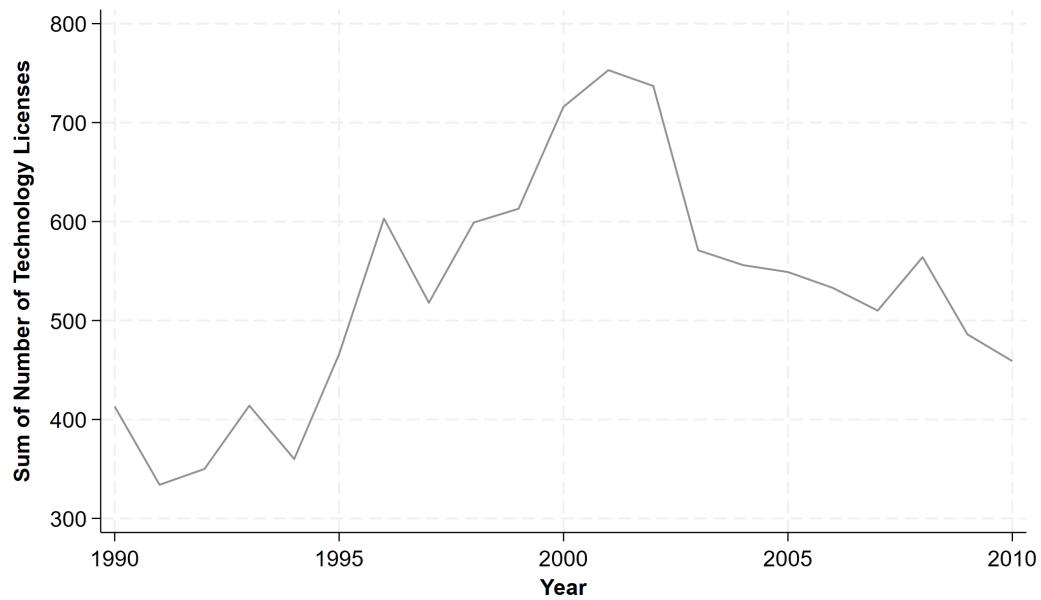
by Gustavo de Souza, Ruben Gaetani, and Martí Mestieri

March 2026

A Summary Statistics

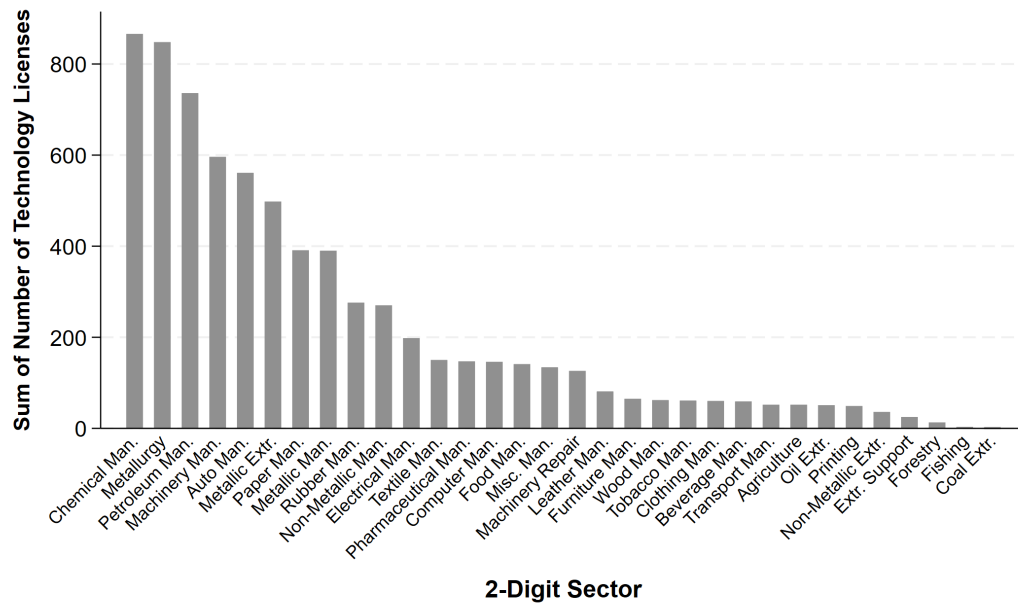
A.1 Summary Statistics of Technology Transfers

Figure A.1: Number of technology transfers over time



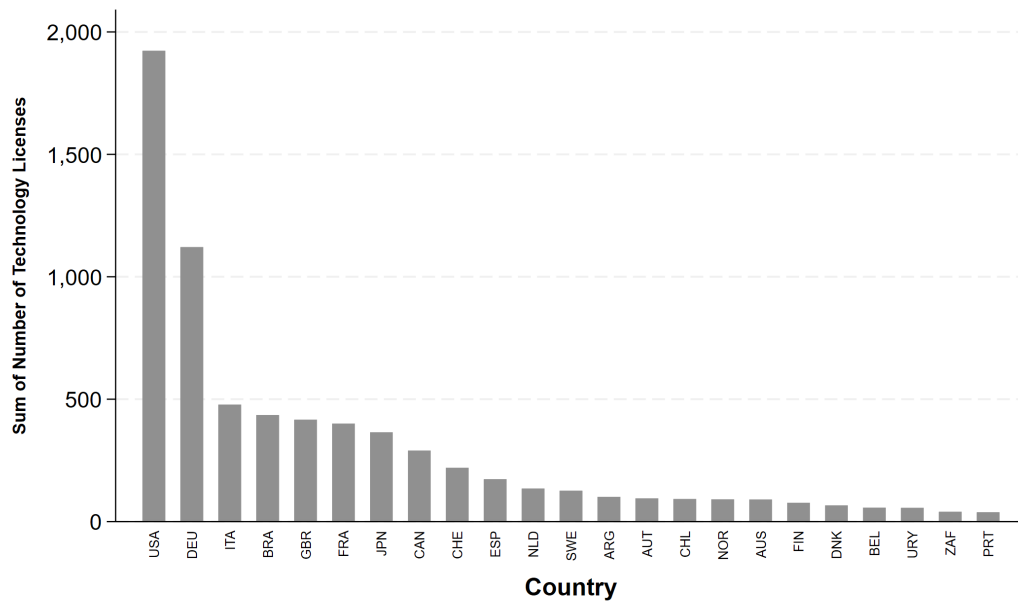
Notes: This figure displays the yearly number of technology transfers received by Brazilian firms.

Figure A.2: Number of technology transfers by sector



Notes: This figure shows the total number of technology transfers received by firms in different sectors.

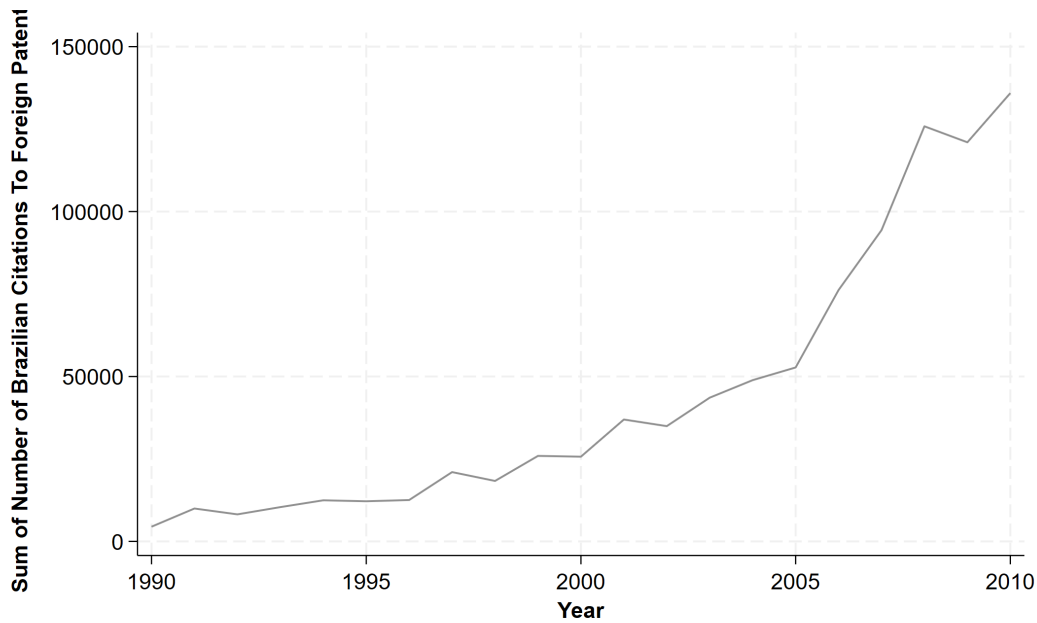
Figure A.3: Number of technology transfers by country of origin



Notes: This figure shows the number of technology transfers made according to the licensor's country of origin. The figure also includes technology transfers made between Brazilian firms.

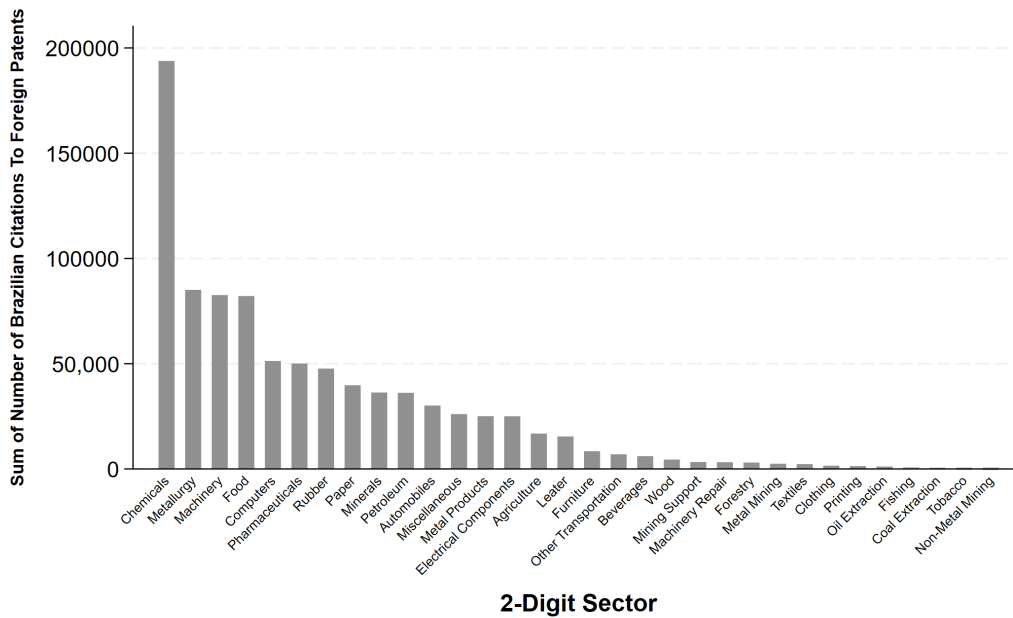
A.2 Summary Statistics of Patent Citations

Figure A.4: Number of citations to foreign patents over time



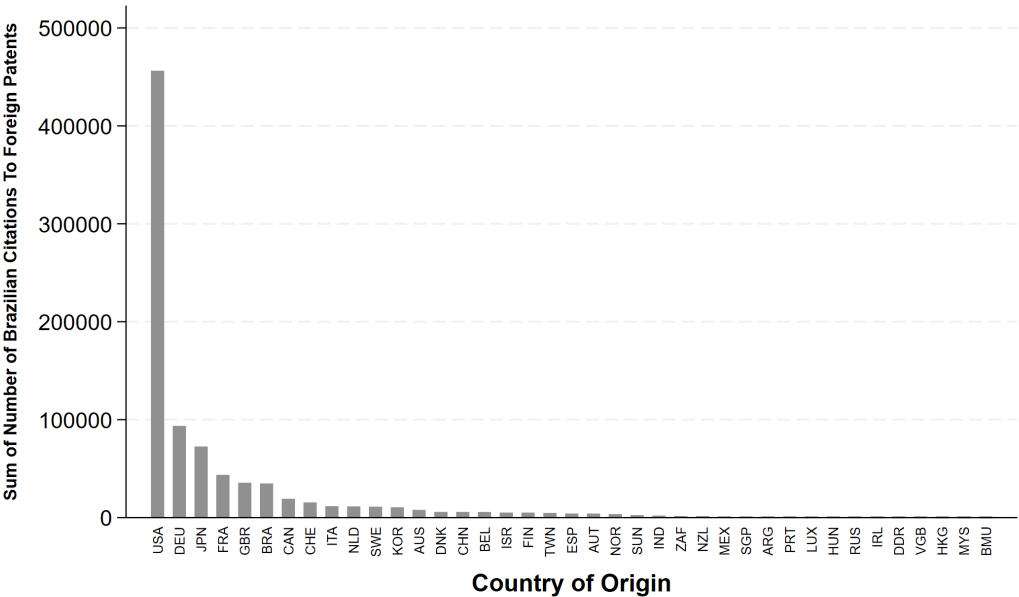
Notes: This figure displays the yearly number of citations made by Brazilian patents.

Figure A.5: Total number of citations to foreign patents by sector



Notes: This figure shows the total number of citations made by Brazilian patents according to the sector of the patent. We link citations to sectors using the crosswalk between International Patent Classification (IPC) fields and sectors created by [Lybbert and Zolas \(2014\)](#).

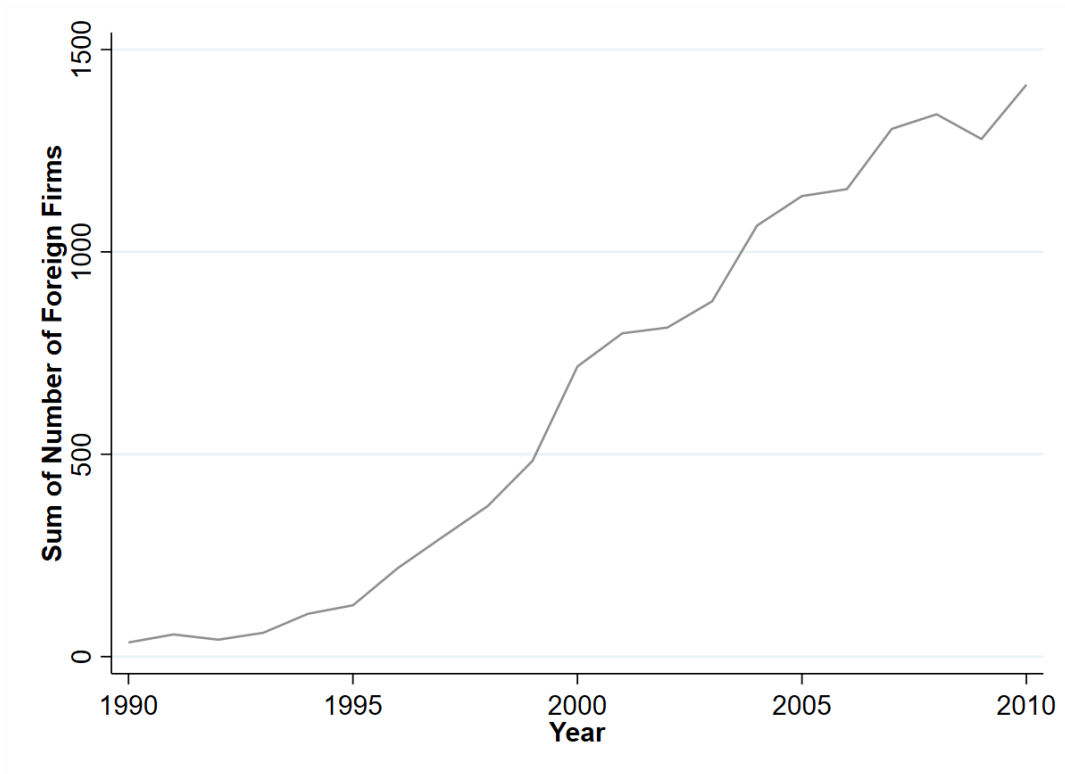
Figure A.6: Total number of citations to foreign patents by country (most important countries)



Notes: This figure shows the total number of citations made by Brazilian patents to patents in foreign countries.

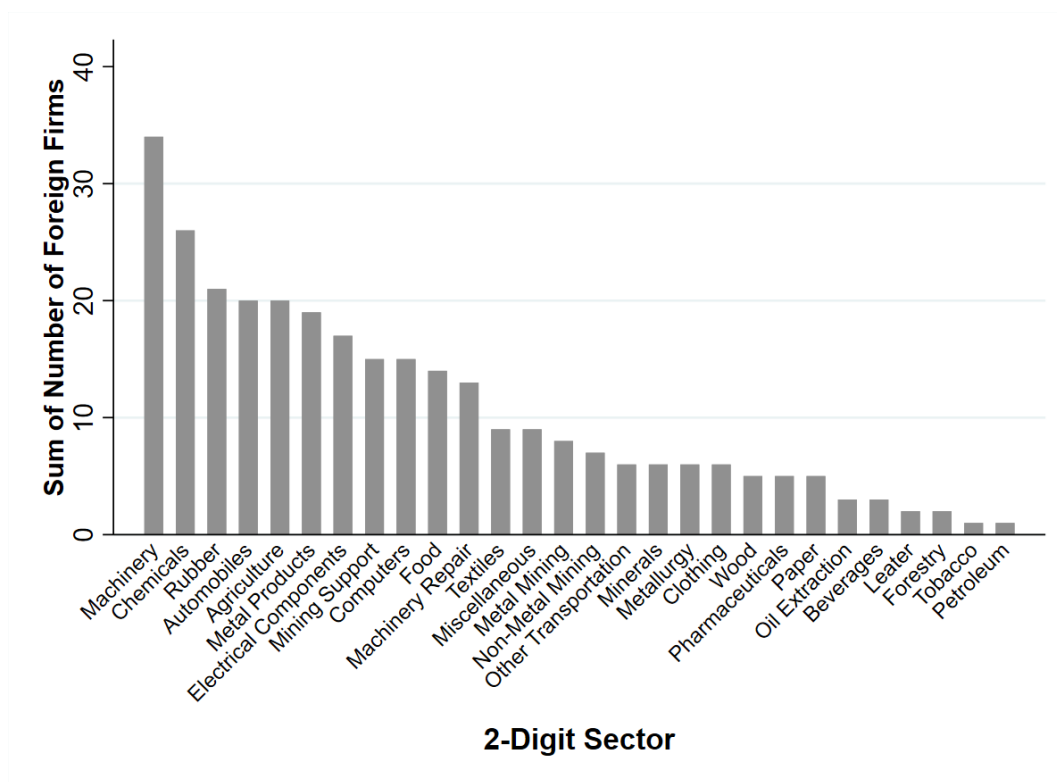
A.3 Summary Statistics of FDI

Figure A.7: Number of firms owned by foreigners over time



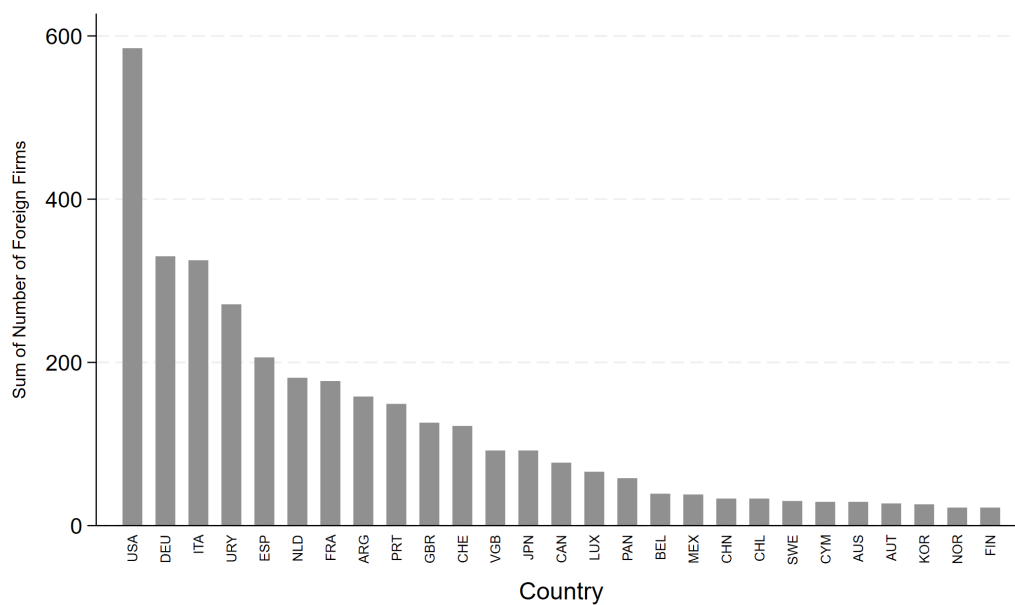
Notes: This figure shows the number of firms with at least one foreign partner per year.

Figure A.8: Number of firms owned by foreigners by sector in 2010



Notes: This figure shows the number of firms with at least one foreign partner per sector in 2010.

Figure A.9: Number of firms owned by foreigners by country in 2010 (most important countries)



Notes: This figure shows the number of firms with at least one foreign partner per country of the foreign partner in 2010.

B Empirics

B.1 Validation

Table B.1: **Instrument, political connections, and outcomes of the origin market**

Panel A: Tariffs and political connections				
	(1)	(2)	(3)	(4)
	<i>Shr Fed.</i>	<i>Shr</i>	<i>log</i>	<i>log Campaign</i>
	<i>Procurement</i>	<i>Donation</i>	<i>Procurement</i>	<i>Contribution</i>
Instrument	-0.0000262 (0.000244)	-0.0000567 (0.000200)	0.0154 (0.0126)	-0.0247 (0.0357)
<i>N</i>	3,146	858	1,103	654
<i>R</i> ²	0.599	0.792	0.837	0.734
Panel B: Tariffs and outcomes of the origin market				
	(1)	(2)	(3)	(4)
	<i>log</i>	<i>Employment</i>	<i>log Value</i>	<i>Value Added</i>
	<i>Employment</i>	<i>Share</i>	<i>Added</i>	<i>Share</i>
Instrument	-1.885 (1.552)	-0.245 (0.218)	2.364 (1.544)	-0.0641 (0.173)
<i>N</i>	3,902	3,902	3,902	3,902
<i>R</i> ²	0.994	0.991	0.996	0.959
Panel C: Tariffs and innovation in other countries				
	(1)	(2)	(3)	(4)
	<i>IHS N.</i>	<i>IHS</i>	<i>IHS N.</i>	<i>IHS</i>
	<i>Cit. by ROW</i>	<i>N. Patents</i>	<i>Cit. by ROW</i>	<i>N. Patents</i>
Instrument	0.0181 (0.157)	0.0262 (0.215)		
Tariff			0.226 (0.149)	0.118 (0.192)
<i>N</i>	1,148,854	1,148,854	1,241,562	1,241,562
<i>R</i> ²	0.086	0.012	0.092	0.014

Notes: Panel A shows the correlation between the instrument and sectoral outcomes in Brazil. The table displays the coefficient of the following regression: $y_{s,t} = \beta \tau_{s,t}^{inst} + \mu_s + \mu_t + \epsilon_{s,t}$, where $y_{s,t}$ is an outcome of sector s in Brazil in year t , $\tau_{s,t}^{inst}$ is the instrument averaged for products of sector s , μ_s is a sector fixed effect, and μ_t is an year fixed effect. The left-hand side variables are the share of firms that received federal procurement (column 1), the share of firms that made a campaign contributions (column 2), the log of total procurement contracts by the federal government (column 3), and the log of total campaign contribution (column 4). Panel B shows the correlation between the instrument and outcomes of the country of origin using the empirical model in Equation (1). The left-hand side variables are log employment (column 1), sectoral employment share (column 2), log value added (column 3), and sectoral value added share (column 4) for the sector in the origin country. Panel C displays estimates of the empirical model in Equation (1), where outcomes are defined as patent citations from the ROW (columns 1 and 3), and as the number of patents in the country of origin-sector affected by Brazilian tariffs (columns 2 and 4). *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table B.2: **Tariff, political connections, and outcomes of the origin market**

Panel A: Tariffs and political connections				
	(1)	(2)	(3)	(4)
	<i>Shr Fed.</i>	<i>Shr</i>	<i>log</i>	<i>log Campaign</i>
	<i>Procurement</i>	<i>Donation</i>	<i>Procurement</i>	<i>Contribution</i>
Tariff	-0.00310	-0.00374	1.478	-2.392
	(0.0235)	(0.0198)	(1.262)	(3.497)
<i>N</i>	3,146	858	1,103	654
<i>R</i> ²	0.599	0.792	0.837	0.734
Panel B: Tariffs and outcomes of the origin market				
	(1)	(2)	(3)	(4)
	<i>log</i>	<i>Employment</i>	<i>log Value</i>	<i>Value Added</i>
	<i>Employment</i>	<i>Share</i>	<i>Added</i>	<i>Share</i>
Tariff	-0.673	-0.127	2.668	-0.0226
	(2.008)	(0.258)	(1.625)	(0.188)
<i>N</i>	4176	4176	4176	4176
<i>R</i> ²	0.994	0.990	0.996	0.958

Notes: Panel A shows the correlation between import tariffs and sectoral outcomes in Brazil. The table displays the coefficient of the following regression: $y_{s,t} = \beta\tau_{s,t} + \mu_s + \mu_t + \epsilon_{s,t}$, where $y_{s,t}$ is an outcome of sector s in Brazil in year t , $\tau_{s,t}$ is the average import tariff against products of sector s , μ_s is a sector fixed effect, and μ_t is an year fixed effect. The left-hand side variables are the share of firms that received a federal procurement (column 1), the share of firms that made a campaign contribution (column 2), the log of total procurement contracts by the federal government (column 3), and the log of total campaign contribution (column 4). Panel B shows the correlation between import tariffs and outcomes of the country of origin using model 1. The left-hand side variables are log employment (column 1), sectoral employment share (column 2), log value added (column 3), and sectoral value added share (column 4) for the sector in the origin country. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

B.2 First Stage

Table B.3: **First Stage**

	(1)	(2)	(3)	(4)
	<i>Tariff</i>	<i>Tariff</i>	<i>Tariff</i>	<i>Tariff</i>
Instrument	0.994*** (0.000192)	0.998*** (0.000125)	0.687*** (0.0121)	0.699*** (0.0123)
Country-Sector FE	N	Y	Y	Y
Country-Year FE	N	N	Y	Y
Sector-Year FE	N	N	Y	Y
Income-Region-Year FE	N	N	N	Y
<i>N</i>	1,392,641	1,392,527	1,392,523	1,148,854
<i>R</i> ²	0.996	0.998	0.999	0.999
F	26,751,637.6	64,037,226.0	3,213.7	3,243.0

Notes: This table presents the first-stage estimates, defined in Equation (2), using different sets of controls. In Column 1, there are no controls. Column 2 adds country-sector fixed effects. Column 3 includes country-year and sector-year fixed effects, while column 4 adds continent-income group-year fixed effects. Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

B.3 Heterogeneous Treatment Effect

We use causal forest to identify the heterogeneity in treatment effects. The goal is to estimate the effect of the tariff conditional on a set of characteristics of the origin country or the home sector in Brazil. In technical terms, we estimate the Conditional Average Treatment Effect (CATE): $E[Y_{1,n,s} - Y_{0,n,s} | Z_{n,s} = z]$, where $Y_{1,n,s}$ and $Y_{0,n,s}$ denote the citations to or technology transfer from country n in sector s with and without a tariff increase, while Z is a set of observable characteristics. Causal forest, as proposed by [Wager and Athey, 2018](#) and [Athey and Imbens, 2019](#), allows for a fully non-parametric relationship between the treatment effect and the set of controls $Z_{n,s}$.

We follow the implementation in [Wager and Athey \(2018\)](#). Because these methods are based in randomized control trials, first we re-write the model in Equation (1) in long-difference (as in [Britto et al., 2022](#)):

$$\Delta y_{n,s} = \beta(Z_{n,s}) \Delta \tau_{n,s}^{inst} + \mu_n^1 + \mu_s^2 + \epsilon_{n,s}, \quad (\text{B.1})$$

where $\Delta y_{n,s}$ is the difference in citations or technology licenses between 2010 and 1990, $\Delta \tau_{n,s}$ is the change in tariffs, μ_n^1 is a country fixed effect, and μ_s^2 is a sector fixed effect. $\beta(Z_{n,s})$ is the treatment effect conditional on variables $Z_{n,s}$. Using the Frisch–Waugh–Lovell theorem ([Frisch and Waugh, 1933](#)), we can re-write as:

$$\Delta \tilde{y}_{n,s} = \beta(Z_{n,s}) \Delta \tilde{\tau}_{n,s}^{inst} + \tilde{\epsilon}_{n,s}$$

where $\Delta \tilde{y}_{n,s}$ is the residual of a regression of the fixed effects, μ_n^1 and μ_s^2 , on $\Delta y_{n,s}$.

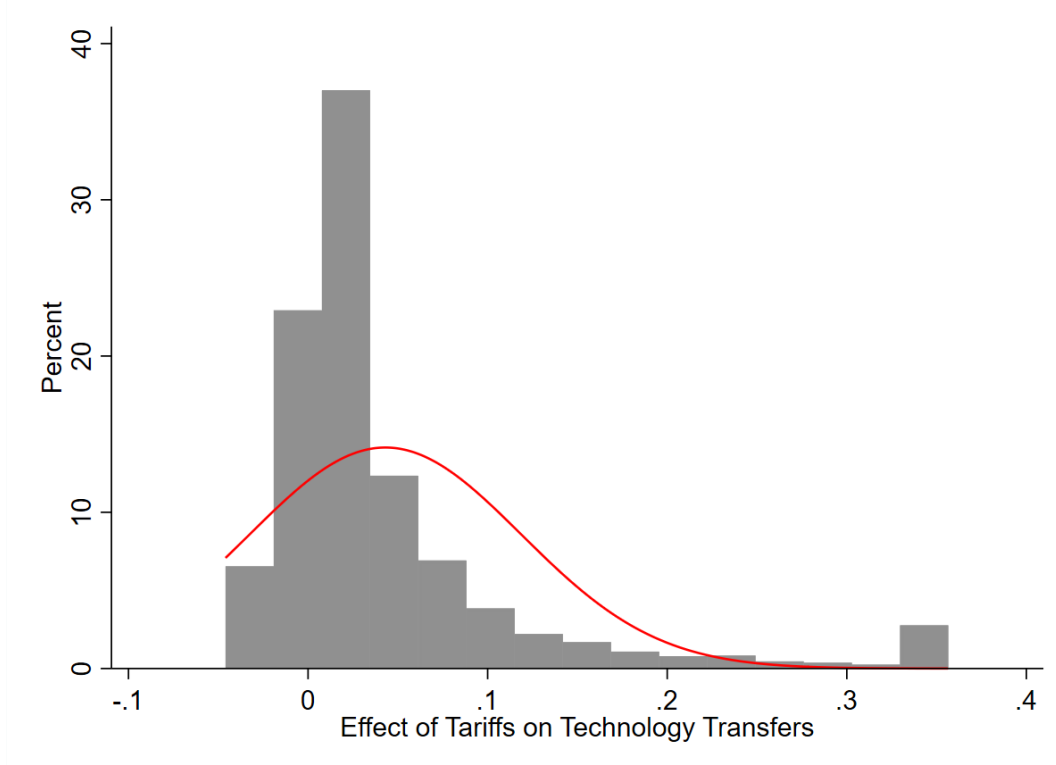
The term $Z_{n,s}$ contains the number of patents issued by country n until 1990, per-capita GDP in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from country n by sector s until 1990, and the number of citations made by sector s to country n until 1990.

As the name suggests, in a causal forest approach, $\beta(Z_{n,s})$ is calculated as the average of several causal trees. Each causal tree is calculated as follows. First, the sample is randomly divided into two groups: one is used to estimate the sample splits (leafs); the other, used for estimation of the CATE, which is curiously called "honest approach". Second, a random set of the covariates $Z_{n,s}$ is selected. Third, the algorithm searches for a split of the sample to maximize the difference in treatment effects in each of the sub-groups, ensuing that in each leaf there are treatments and controls. The more a variable is used to split the sample, larger is its importance

to predict heterogeneity in the treatment effect. Forth, the process continues until the leaf or the heterogeneity in treatment effects between leafs is too small. This process is repeated 10,000 times and averaged out on the estimation sample.

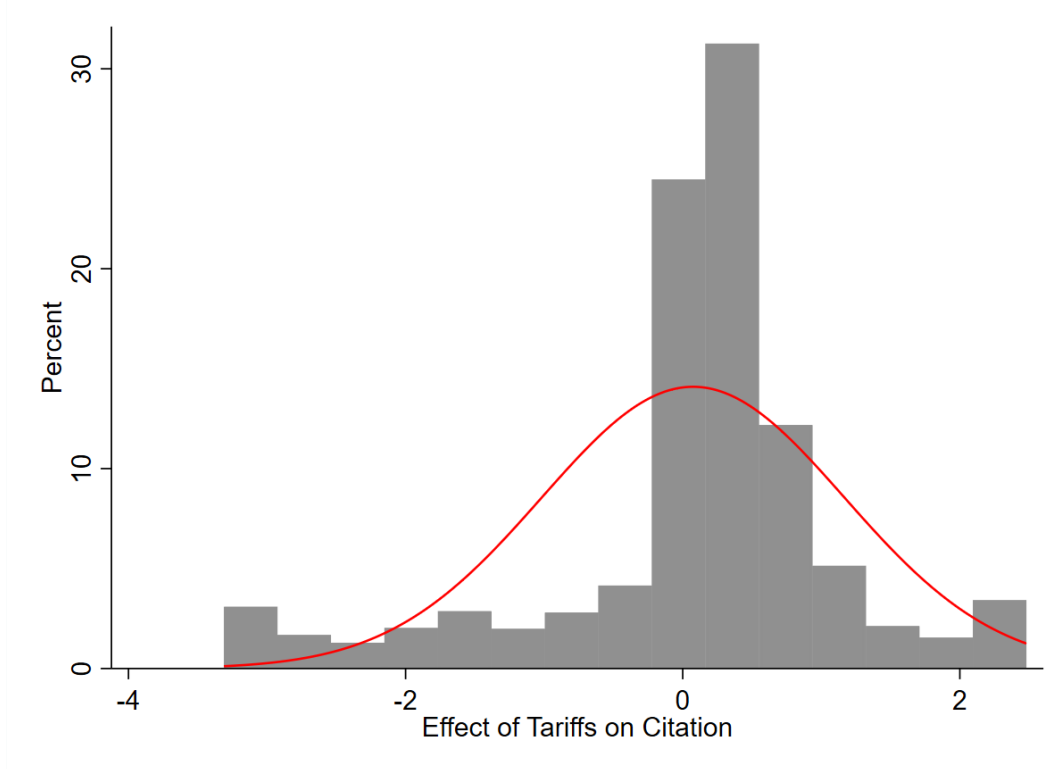
Distribution of treatment effect. Figure B.11 shows that there is a large degree of heterogeneity of the effect of tariffs on technology transfers and citations.

Figure B.10: **Distribution of the CATE of tariffs on technology transfers**



Notes: This figure shows the histogram of the conditional average treatment effect, $\beta(Z_{n,s})$, capturing how tariffs affect technology transfers. The covariates include the number of patents issued by country n until 1990, per-capita GDP in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from country n by sector s until 1990, and the number of citations made by sector s to country n until 1990.

Figure B.11: **Distribution of the CATE of tariffs on citations**



Notes: This figure shows the histogram of the conditional average treatment effect, $\beta(Z_{n,s})$, capturing how tariffs affect citations. The covariates include the number of patents issued by country n until 1990, per-capita GDP in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from country n by sector s until 1990, and the number of citations made by sector s to country n until 1990.

Correlation of treatment effects. Tables B.4 and B.5 show the correlation of different observables of the market of origin with the CATE of tariffs on technology transfer or citations. The effect of tariffs on technology transfer and citations is larger when the origin country has larger GDP and patent stock. The effect is also larger if the Brazilian sector is highly innovative or has a history of technology transfers.

Heterogeneity by sector. Figures B.12 and B.13 shows the average CATE by sector. The extractive sectors, chemicals, and automobile manufacturing are the ones most affected.

Heterogeneity by Country. Figure B.14 and B.15 shows the countries with largest CATE.

Table B.4: **Correlation between Effect of Tariff on Technology Transfers and Pre-Period Characteristics**

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$\beta_{tech.transf.}$	$\beta_{tech.transf.}$	$\beta_{tech.transf.}$	$\beta_{tech.transf.}$	$\beta_{tech.transf.}$	$\beta_{tech.transf.}$	$\beta_{tech.transf.}$
$\beta_{citation}$	0.0144*** (0.000415)						
$\log(GDP\ Per\ Capita)$		0.0144*** (0.000379)					0.0152*** (0.000433)
$IHS(Patents\ Origin)$			0.00208*** (0.000206)				-0.00091*** (0.000280)
$IHS(Patents\ BR)$				-0.00430*** (0.000244)			-0.00575*** (0.000290)
$IHS(Citations)$					0.0104*** (0.00239)		-0.00780*** (0.00271)
$IHS(Tech.\ Transf.)$						0.0487*** (0.00597)	0.0317*** (0.00616)
N	54,379	41,862	54,379	54,379	54,379	54,379	41,862
R^2	0.022	0.033	0.002	0.006	0.000	0.001	0.045

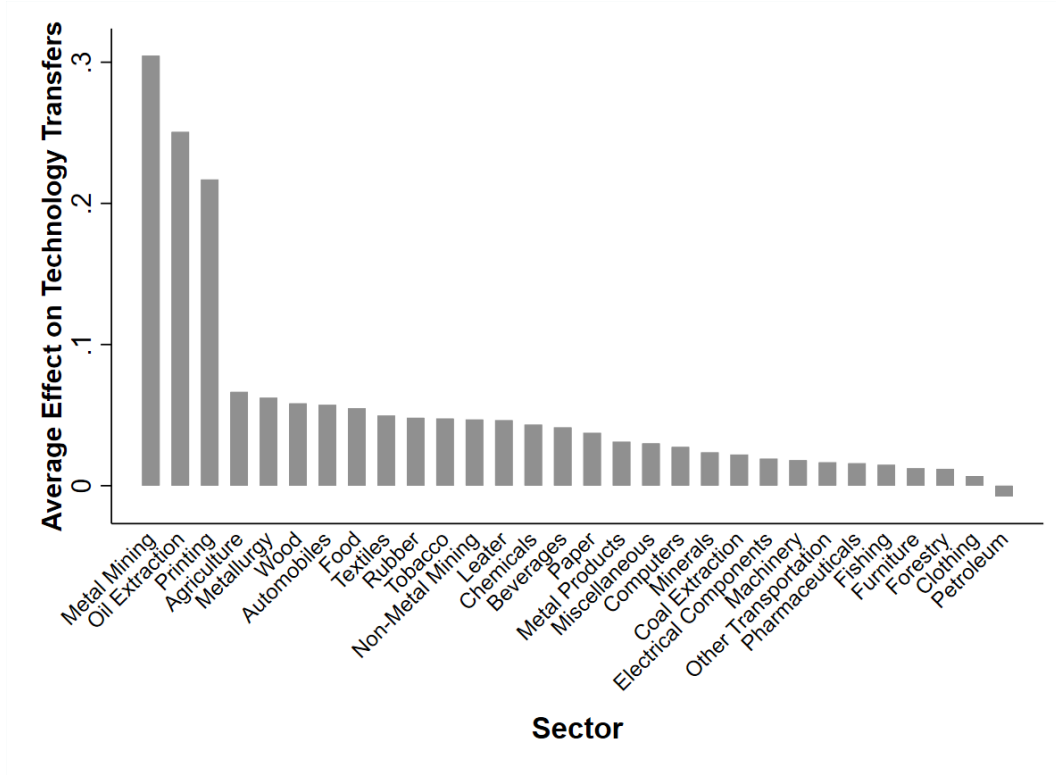
Notes: This table presents the correlation between the effect of tariff on technology transfers and pre-period characteristics. Each column represents a different specification where the dependent variable is the effect of tariffs on technology transfers. Standard errors are in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table B.5: **Correlation between Effect of Tariff on Citations and Pre-Period Characteristics**

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$\beta_{citation}$	$\beta_{citation}$	$\beta_{citation}$	$\beta_{citation}$	$\beta_{citation}$	$\beta_{citation}$	$\beta_{citation}$
$\beta_{tech.transf.}$	1.503*** (0.0434)						
$\log(GDP\ Per\ Capita)$		-0.0690*** (0.00157)					-0.0930*** (0.00170)
$IHS(Patents\ Origin)$			0.0286*** (0.00211)				0.0336*** (0.00110)
$IHS(Patents\ BR)$				0.00942*** (0.00250)			0.0582*** (0.00114)
$IHS(Citations)$					0.147*** (0.0244)		-0.0640*** (0.0107)
$IHS(Tech.\ Transf.)$						0.0392 (0.0611)	-0.0702*** (0.0243)
N	54,379	41,862	54,379	54,379	54,379	54,379	41,862
R^2	0.022	0.044	0.003	0.000	0.001	0.000	0.142

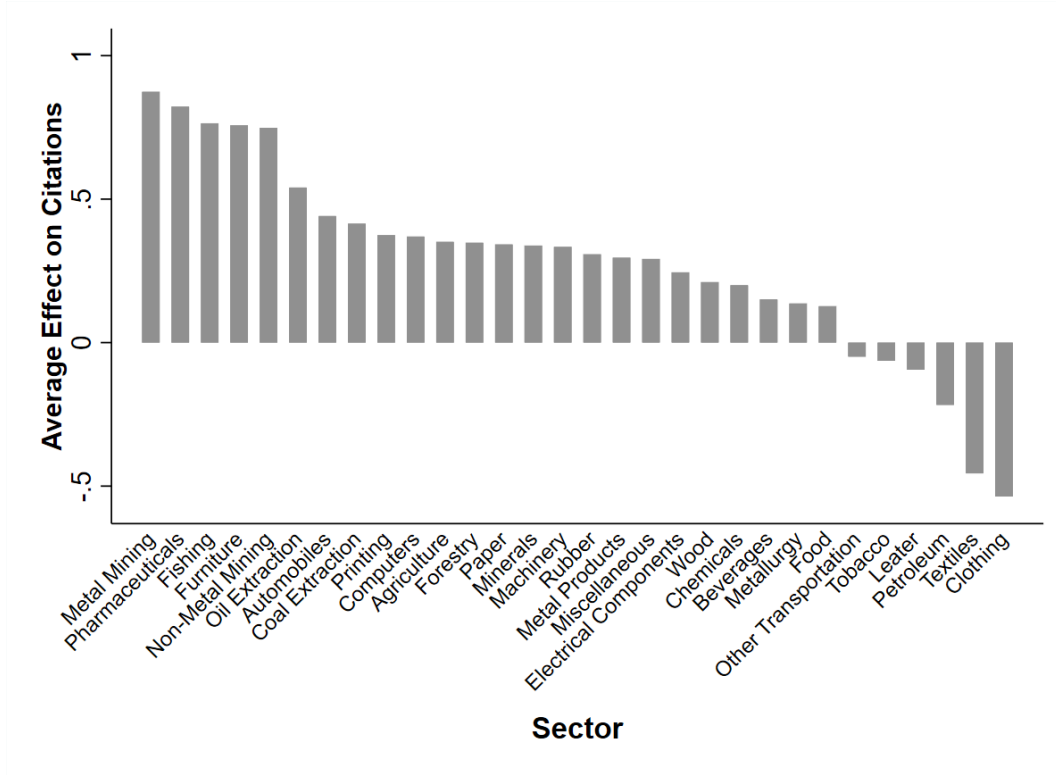
Notes: This table presents the correlation between the effect of tariff on citations and pre-period characteristics. Each column represents a different specification where the dependent variable is the effect of tariffs on citations. Standard errors are in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Figure B.12: Effect of tariffs on technology transfers by sector



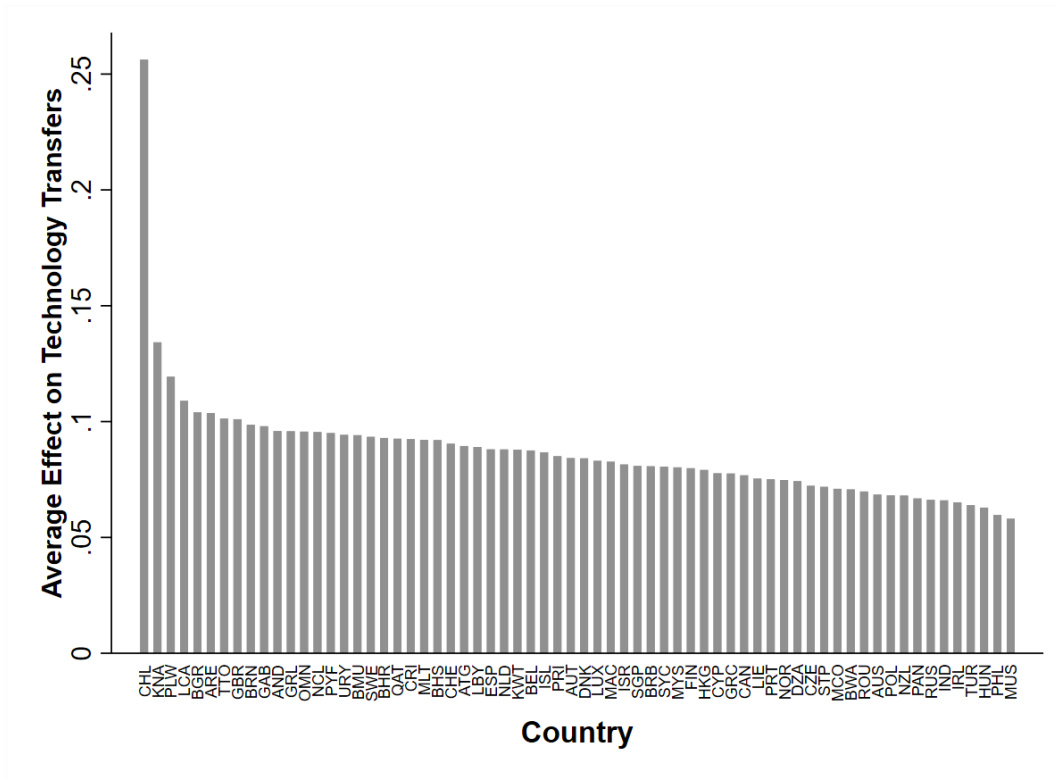
Notes: This figure shows the average of the conditional average treatment effect, $\beta(Z_{n,s})$, capturing how tariffs affect technology transfers by sector. For each sector we calculate $E_s[\beta(Z_{n,s})]$. The covariates include the number of patents issued by country n until 1990, per-capita GDP in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from country n by sector s until 1990, and the number of citations made by sector s to country n until 1990.

Figure B.13: Effect of tariffs on citations by sector



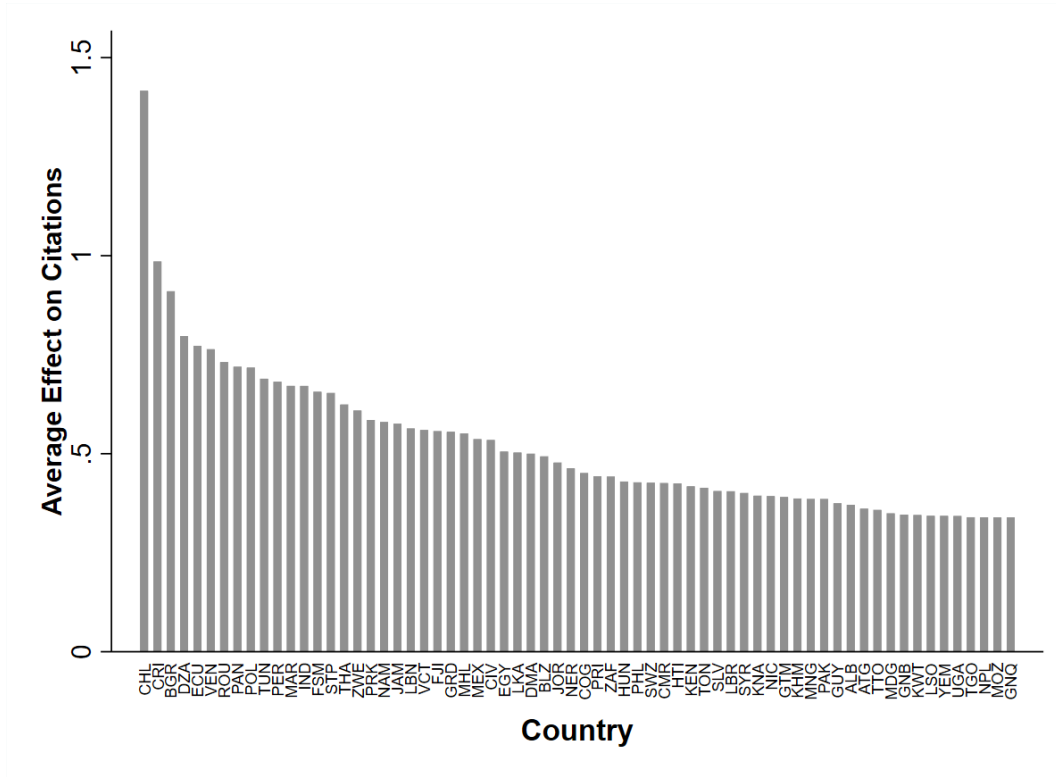
Notes: This figure shows the average of the conditional average treatment effect, $\beta(Z_{n,s})$, capturing how tariffs affect citations by sector. For each sector we calculate $E_s[\beta(Z_{n,s})]$. The covariates include the number of patents issued by country n until 1990, per-capita GDP in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from country n by sector s until 1990, and the number of citations made by sector s to country n until 1990.

Figure B.14: Effect of tariffs on technology transfers by country



Notes: This figure shows the average of the conditional average treatment effect, $\beta(Z_{n,s})$, capturing how tariffs affect technology transfers by country. For each country we calculate $E_n[\beta(Z_{n,s})]$. The covariates include the number of patents issued by country c until 1990, per-capita GDP in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from country n by sector s until 1990, and the number of citations made by sector s to country n until 1990.

Figure B.15: Effect of tariffs on citation by country



Notes: This figure shows the average of the conditional average treatment effect, $\beta(Z_{n,s})$, capturing how tariffs affect citation by country. For each country we calculate $E_n[\beta(Z_{n,s})]$. The covariates include the number of patents issued by country n until 1990, per-capita GDP in 1990, the number of patents issued by sector s in Brazil until 1990, the number of technology transfers from country n by sector s until 1990, and the number of citations made by sector s to country n until 1990.

B.4 OLS Regressions

Table B.6: **Import tariffs and technology diffusion with OLS regressions**

	(1)	(2)	(3)	(4)
	<i>IHS N. Tech. Transfers</i>	<i>IHS Citations</i>	<i>IHS. Cit. to Non-Licensors</i>	<i>IHS Cit. to Licensor</i>
Tariff	0.0950** (0.0393)	0.416*** (0.0673)	-0.0314 (0.0379)	0.468*** (0.0594)
<i>N</i>	1,271,697	1,271,697	1,271,697	1,271,697
<i>R</i> ²	0.000	0.039	0.012	0.049

Notes: This table reports the coefficients of regressing different measures of technology transfers to Brazil on tariffs, according to the model in Equation (1) without using any instrument. All specifications use the inverse hyperbolic sine transformation. The left-hand side variable is the number of technology transfers (column 1), the number of citations (column 2), the number of citations to firms that never sent technology to Brazil (column 3), and the number of citations to patents of firms sending technology to Brazil or that are cited by them (column 4). Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

B.5 Controls

Table B.7: **Import tariffs and technology diffusion under different controls**

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>IHS</i>	<i>IHS</i>	<i>IHS</i>	<i>IHS</i>	<i>IHS</i>	<i>IHS</i>
	<i>N.Tech.</i>	<i>Citations</i>	<i>N.Tech.</i>	<i>Citations</i>	<i>N.Tech.</i>	<i>Citations</i>
Tariff	0.105** (0.0483)	0.651*** (0.118)	0.143** (0.0592)	0.781*** (0.125)	0.107** (0.0486)	0.610*** (0.108)
Income-Region FE	N	N	Y	Y	N	N
Lagged LHS	N	N	N	N	Y	Y
<i>N</i>	1,392,523	1,392,523	1,148,854	1,148,854	1,392,523	1,392,523

Notes: This table reports the coefficients of regressing different measures of technology diffusion on tariffs, according to the model in Equation (1) using the first-stage regression 2. All specifications use the inverse hyperbolic sine transformation. Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

B.6 Sample Selection

Table B.8: **Import tariffs and technology diffusion constraining sample to high-income countries**

	(1)	(2)	(3)	(4)
	<i>IHS Transfers</i>	<i>IHS Citations</i>	<i>IHS Cit. to Non- Licensor</i>	<i>IHS Cit. to Licensor</i>
Tariffs	0.489** (0.211)	0.852*** (0.218)	0.104** (0.0466)	0.739*** (0.219)
<i>N</i>	381,204	381,204	381,204	381,204

Notes: This table reports the coefficients of regressing different measures of technology diffusion on tariffs, according to the model in Equation (1) using the first-stage regression 2. All specifications use the inverse hyperbolic sine transformation. The sample is limited to high-income countries according to the world bank definition. Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

B.7 Alternative Functional Forms

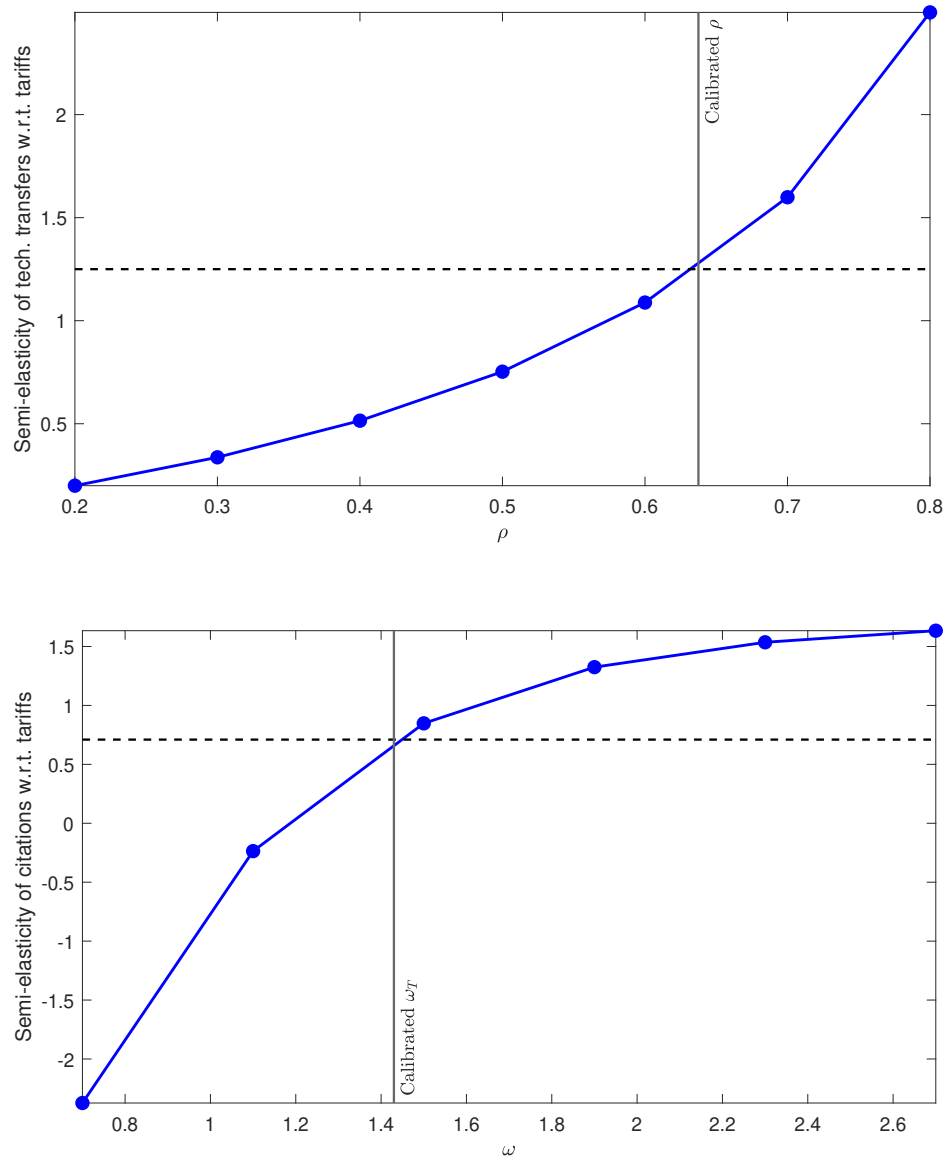
Table B.9: Effect of tariffs on international technology licensing

	(1) <i>At Least One Tech. Transf.</i>	(2) <i>At Least One Citation</i>	(3) <i>Percentile of N. Tech. Transf.</i>	(4) <i>Percentile of Citations</i>	(5) <i>N. Tech. Transf.</i>	(6) <i>Citations</i>
Tariffs	0.0847*** (0.0301)	0.539*** (0.0838)	8.382*** (2.976)	51.62*** (8.011)	0.229** (0.0953)	0.887*** (0.190)
<i>N</i>	1,178,000	1,178,000	1,178,000	1,178,000	1,178,000	1,178,000

Notes: This table reports the coefficients of regressing different measures of technology diffusion on tariffs, according to the model in Equation (1) using the first-stage regression 2. Column 1 shows the effect of tariffs on an indicator taking value of one for at least one technology transfer, column 2 on an indicator for at least one citation, column 3 on the percentile of technology transfer, column 4 on the percentile of citations, column 5 on the number of technology transfers, and column 6 on the number of citations. Standard errors are clustered at the country-sector level. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

C Additional model figures

Figure C.16: Identification of ρ and ω_L



Notes: Semi-elasticities for a range of values of ρ (top panel) and ω_T (bottom panel). All other parameters are kept constant.

D Derivations

In this section, we provide details on the derivations of the equilibrium conditions and results of the model presented in Section 5.

D.1 Prices and Sectoral Demand

Letting C_t^n be final consumption in country n at time t , and letting $P_t^{n,ks}$ be the price aggregator of sector s , total demand for sector s satisfies

$$C_t^{n,s} = \left(\frac{\mathcal{P}_t^n}{P_t^{n,s}} \right)^\sigma C_t^n, \quad (\text{D.2})$$

where $C_t^{n,s} = \left(\int_0^1 c_t^{n,s}(i)^{\frac{\epsilon-1}{\epsilon}} di \right)^{\frac{\epsilon}{\epsilon-1}}$. The price aggregator of final consumption, $\mathcal{P}_t^n$, defined as $\mathcal{P}_t^n C_t^n = \sum_s C_t^{n,s} P_t^{n,s}$, is equal to

$$\mathcal{P}_t^n = \left[\sum_s (P_t^{n,s})^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (\text{D.3})$$

Similarly, demand of variety i in sector s satisfies

$$c_t^{n,s}(i) = \left(\frac{P_t^{n,s}}{p_t^{n,s}(i)} \right)^\epsilon C_t^{n,s}, \quad (\text{D.4})$$

while the sectoral price aggregator is equal to

$$P_t^{n,s} = \left(\int_0^1 (p_t^{n,s}(i))^{1-\epsilon} di \right)^{\frac{1}{1-\epsilon}}. \quad (\text{D.5})$$

The assumptions on two-stage pricing imply that each variety in each country will only be supplied by the producer with the lowest marginal cost, that will charge the optimal markup under monopolistic competition. Since the derivation of the price level and expenditure shares in foreign countries are otherwise identical to the standard Eaton-Kortum model, in what follows we focus on the derivations for Brazil (B).

For each variety i , the price faced by consumers in country B is equal to

$$p_t^{B,s}(i) = \frac{\epsilon}{\epsilon - 1} \times \min \left\{ \frac{w_t^B}{q_t^{B,s}(i)}, \left\{ \frac{w_t^B d_t^{n,s}}{q_{T,t}^{n,s}(i)}, \frac{w_t^n \kappa_t^{n \rightarrow B,s}}{q_{X,t}^{n,s}(i)} \right\}_{n=1,\dots,N} \right\}, \quad (\text{D.6})$$

The cumulative distribution function of prices in sector s in country B is

$$Pr \left\{ p_t^{B,s}(i) \leq p \right\} = Pr \left\{ \min \{ p_t^{n \rightarrow B,s}(i) \leq p \} \right\} = 1 - Pr \left\{ \frac{\epsilon w_{k,t}^{n \rightarrow B} \bar{\kappa}_{k,t}^{n \rightarrow B,s}}{(\epsilon - 1) q_{k,t}^{n,s}(i)} > p, \forall (n, k) \right\}, \quad (\text{D.7})$$

where $k \in \{X, T\}$, $w_{k,t}^{n \rightarrow B}$ denotes the relevant wage that applies to source (n, k) in Brazil,²⁹ and $\bar{\kappa}_t^{(n,k) \rightarrow B,s}$ is equal to $\kappa_t^{n \rightarrow B,s}$ if $k = X$ and is equal to $d_t^{n,s}$ if $k = T$. Rearranging:

$$Pr \left\{ p_t^{B,s}(i) \leq p \right\} = 1 - Pr \left\{ \frac{(\epsilon - 1) q_{k,t}^{n,s}(i)}{\epsilon w_{k,t}^{n \rightarrow B} \bar{\kappa}_{k,t}^{n \rightarrow B,s}} < \frac{1}{p}, \forall (n, k) \right\} = 1 - \exp \left\{ -p^{-\theta} \left(\frac{\epsilon}{\epsilon - 1} \right)^{-\theta} G_t^{B,s} \right\}, \quad (\text{D.8})$$

where $G_t^{B,s}$ is defined as:

$$G_t^{B,s} \equiv G^B \left(\lambda_t^{B,s} (w_t^B)^{-\theta}, \left\{ \lambda_t^{n,s} (w_t^B d_t^{n,s})^{-\theta}, \lambda_t^{n,s} (w_t^n \kappa_t^{n \rightarrow B,s})^{-\theta} \right\}_{n=1,\dots,N} \right). \quad (\text{D.9})$$

In Equation (D.9), $G^B(\cdot)$ denotes the correlation function across the different source productivity distributions, which in this case is defined as

$$G^B(x^B, \{x_T^n, x_X^n\}_{n=1,\dots,N}) \equiv x_B + \sum_{n=1}^N \left[(x_T^n)^{\frac{1}{1-\rho}} + (x_X^n)^{\frac{1}{1-\rho}} \right]^{1-\rho}. \quad (\text{D.10})$$

Integrating over all prices with respect to the distribution in Equation (D.8), we obtain the price index for sector k :

$$P_t^{B,s} = \frac{\epsilon}{\epsilon - 1} (G_t^{B,s})^{-\frac{1}{\theta}} \Gamma \left(\frac{1 - \epsilon + \theta}{\theta} \right)^{\frac{1}{1-\epsilon}}, \quad (\text{D.11})$$

where Γ denotes the Gamma function. This expression can then be plugged into Equation (D.3) to obtain the overall price index.

Finally, the share of varieties consumed in B that originate from (n, k) (which, due to the standard properties of the Eaton-Kortum model, also corresponds to the expenditure share from

²⁹This wage is equal to w_t^n if $k = X$ and is equal to w_t^B if $k = T$ or $n = B$.

B to (n, k) is equal to:

$$\pi_{k,t}^{n \rightarrow B,s} = \frac{\lambda_t^{n,s} (\tilde{w}_{k,t}^{n \rightarrow B} \bar{\kappa}_{k,t}^{n \rightarrow B,s})^{-\theta} \frac{\partial G_t^{B,s}}{\partial(n,k)}}{G_t^{B,s}}, \quad (\text{D.12})$$

where $\frac{\partial G_t^{B,s}}{\partial(n,k)}$ denotes the partial derivative of $G_t^{B,s}$ with respect to x_k^n .

The derivation for foreign countries is analogous to the one for Brazil, with the exception that varieties in those countries cannot be provided via a technology transfer. Hence, the price faced for variety i in sector s by the representative consumer in country n is given by

$$p_t^{n,s}(i) = \frac{\epsilon}{\epsilon - 1} \times \min \left\{ \frac{w_t^m \kappa_t^{m \rightarrow n,s}}{q_{X,t}^{m,s}(i)} \right\}_{m=B,1,\dots,N}. \quad (\text{D.13})$$

Since, in this case, all source distributions are independent, expenditure shares can be written as

$$\pi_t^{m \rightarrow n,s} = \frac{\lambda_t^m (w_t^m \kappa_t^{m \rightarrow n,s})^{-\theta}}{G_t^{m,s}}, \quad (\text{D.14})$$

where

$$G_t^{n,s} \equiv \sum_{m=B,1,\dots,N} \lambda_t^{m,s} (w_t^m \kappa_t^{m \rightarrow n,s})^{-\theta}.$$

Analogous expressions can be obtained for sectoral price indexes ($P_t^{n,s}$) and the overall price aggregator ($\mathcal{P}_t^n$).

D.2 Evolution of the productivity distribution in Brazil

The process described by Equations (14) and (15) gives rise to the following law of motion for the cumulative distribution function (CDF) of productivity of local entrepreneurs in Brazil, $F_t^{B,s}(q')$:

$$1 - F_{t+\Delta}^{B,s}(q') = [1 - F_t^{B,s}(q')] + F_t^{B,s}(q') \int_t^{t+\Delta} \sum_{n=1,\dots,N} \sum_{k=X,T} \left[\int A_{k,\tilde{t}}^{n \rightarrow B,s} \left(\frac{q'}{\zeta_t^{n,s} \omega_k(q)^\beta} \right) dF_{k,\tilde{t}}^{n \rightarrow B,s}(q) \right] d\tilde{t},$$

where $F_{k,t}^{n \rightarrow B,s}$ is the distribution of productivity of *active* foreign exporters ($k = X$) or providers of technology ($k = T$), respectively, from country n selling their products in B .

Rearranging, taking the limit for $\Delta \rightarrow 0$, and using the definition of $A_{k,t}^{n \rightarrow B,s}(z)$ in Equation

(14), we obtain the following differential equation describing the dynamics of the local CDF:

$$\frac{d \ln F_t^{B,s}(q')}{dt} = -(q')^{-\theta} \Lambda_t^{B,s}, \quad (\text{D.15})$$

where

$$\Lambda_t^{B,s} \equiv \alpha_t^s \sum_{n=1,\dots,N} (\zeta_t^{n,s})^\theta \left[\pi_{X,t}^{n \rightarrow B,s} \int_0^\infty x^{\beta\theta} dF_{X,t}^{n \rightarrow B,s}(x) + (\omega_T)^\theta \pi_{T,t}^{n \rightarrow B,s} \int_0^\infty x^{\beta\theta} dF_{T,t}^{n \rightarrow B,s}(x) \right], \quad (\text{D.16})$$

where we have already imposed $\omega_X = 1$. Note that the right-hand-side is equivalent to the right-hand-side of Equation (16).

Solving the differential equation in (D.15), we obtain

$$F_t^{B,s}(q') = F_0^{B,s}(q') \exp \left(-(q')^{-\theta} \int_0^t \Lambda_{\tilde{t}}^{B,s} d\tilde{t} \right). \quad (\text{D.17})$$

Using Equation (11) to replace $F_0^{B,s}(q)$ and letting $\lambda_t^{B,s} = \lambda_0^{B,s} + \int_0^t \Lambda_{\tilde{t}}^{B,s} d\tilde{t}$. This expression verifies that the CDF of best productivities among B producers is indeed Fréchet with shape parameter θ and with scale parameter $\lambda_t^{B,s}$, whose law of motion satisfies

$$\dot{\lambda}_t^{B,s} = \Lambda_t^{B,s}, \quad (\text{D.18})$$

implying Equation (16).

It is straightforward to show that

$$\int_0^\infty x^{\beta\theta} dF_{k,t}^{n \rightarrow B,s}(x) = \Gamma(1 - \beta) \left(\frac{\lambda_t^{n,s} \frac{\partial G_t^{B,s}}{\partial(n,k)}}{\pi_{k,t}^{n \rightarrow B,s}} \right)^\beta. \quad (\text{D.19})$$

Hence, the law of motion for $\lambda_t^{B,s}$ can be written as

$$\dot{\lambda}_t^{B,s} \propto \alpha_t^s \sum_{n=1,\dots,N} (\zeta_t^{n,s})^\theta \left[\pi_{X,t}^{n \rightarrow B,s} \left(\frac{\lambda_t^{n,s} \frac{\partial G_t^{B,s}}{\partial(n,X)}}{\pi_{X,t}^{n \rightarrow B,s}} \right)^\beta + (\omega_T)^\theta \pi_{T,t}^{n \rightarrow B,s} \left(\frac{\lambda_t^{n,s} \frac{\partial G_t^{B,s}}{\partial(n,T)}}{\pi_{T,t}^{n \rightarrow B,s}} \right)^\beta \right]. \quad (\text{D.20})$$

D.3 Details on the mapping between the model and the empirical results

D.3.1 Derivation of Proposition 1 and interpretation of $f_T(\cdot)$ and ε_T

We first show that the value $V_{T,t}^{n,s}$, defined as the total payments from Brazil to foreign providers of technology transfers, follows the gravity representation in Equation (17). Then, we provide and discuss close-form expressions for $f_T(\cdot)$ and ε_T .

To derive Equation (17), start from the fact that $V_{T,t}^{n \rightarrow B,s}$ can be expressed as a sector-time-specific term, $\bar{v}_t^s$, multiplied by the share of sector- s varieties sold in Brazil produced with technology transfers from country n :

$$V_{T,t}^{n \rightarrow B,s} = \bar{v}_{s,t} \pi_{T,t}^{n \rightarrow B,s}.$$

From Equation (D.12), we know that the share $\pi_{T,t}^{n \rightarrow B,s}$ can be expressed as

$$\pi_{T,t}^{n \rightarrow B,s} = \frac{\lambda_t^{n,s} (w_t^B d_t^{n,s})^{-\theta} g_{T,t}^{n,s}}{G_t^{B,s}}, \quad (\text{D.21})$$

where $G_t^{B,s}$ is the sector-time-specific correlation function, and $g_{T,t}^{n,s}$ is its partial derivative with respect to the entry corresponding to technology transfers from n in sector s :

$$g_{T,t}^{n,s} \equiv \frac{\partial G_t^{B,s}}{\partial(n, T)} = \left[1 + \left(\frac{w_t^n \bar{\kappa}^{n \rightarrow B,s} (1 + \tau_t^{n \rightarrow B,s})}{w_t^B d_t^{n,s}} \right)^{\frac{-\theta}{1-\rho}} \right]^{-\rho}. \quad (\text{D.22})$$

Taking logs of Equation (D.21) and taking a first-order approximation around a baseline period $t = \bar{t}$, for any arbitrary perturbation in productivity ($\lambda_t^{n,s}$), transfer costs ($d_t^{n,s}$), tariffs ($\tau_t^{n \rightarrow B,s}$), and resulting changes in relative wages (w_t^n/w_t^B), we obtain the following expression:

$$\log(V_{T,t}^{n \rightarrow B,s}) \approx \mu_{n,s}^1 + \mu_{n,t}^2 + \mu_{s,t}^3 + \mathbf{E}_{m,r} \left[\frac{\partial \log g_{T,t'}^{m,r}}{\partial \tau_{t'}^{m \rightarrow B,r}} \bigg|_{t'=\bar{t}} \right] \tau_t^{n \rightarrow B,s} + \varepsilon_{T,t}^{n,s}, \quad (\text{D.23})$$

where the term $\varepsilon_{T,t}^{n,s}$ has mean zero and all terms that only vary at the level of the country-sector, country-time, and sector-time, are grouped in their respective fixed effect, $\mu_{\cdot,\cdot}^i$.

The coefficient of the gravity equation, then, corresponds to

$$\begin{aligned} \text{Structural coeff.} \\ \text{(tech. transfers gravity)} \end{aligned} = \mathbf{E}_{m,r} \left[\frac{\partial \log g_{T,t'}^{m,r}}{\partial \tau_{t'}^{m \rightarrow B,r}} \bigg|_{t'=\bar{t}} \right] \quad (\text{D.24})$$

that is, the average across countries (m) and sectors (s) of their response to a change in their respective tariffs, $\tau_t^{m \rightarrow B, r}$.

The country-sector-specific response behind this average treatment effect has a closed-form expression:

$$\left. \frac{\partial \log g_{T, t'}^{n, s}}{\partial \tau_t^{n \rightarrow B, s}} \right|_{t'=\bar{t}} = \frac{\theta \rho}{1 - \rho} \frac{\left(\frac{(w_{\bar{t}}^n \bar{\kappa}^{n \rightarrow B, s} (1 + \tau_{\bar{t}}^{n \rightarrow B, s}))^{-\theta}}{(w_{\bar{t}}^B d_{\bar{t}}^{n, s})^{-\theta}} \right)^{\frac{1}{1-\rho}}}{\left[1 + \left(\frac{(w_{\bar{t}}^n \bar{\kappa}^{n \rightarrow B, s} (1 + \tau_{\bar{t}}^{n \rightarrow B, s}))^{-\theta}}{(w_{\bar{t}}^B d_{\bar{t}}^{n, s})^{-\theta}} \right)^{\frac{1}{1-\rho}} \right]} (1 + \tau_{\bar{t}}^{n \rightarrow B, s})^{-1}. \quad (\text{D.25})$$

Conditional on observable $(w_{\bar{t}}^n/w_{\bar{t}}^B$ and $\tau_{\bar{t}}^{n \rightarrow B, s})$ and invertible $(\bar{\kappa}^{n \rightarrow B, s}$ and $d_{\bar{t}}^{n, s})$ variables at the baseline period $\bar{t}$, and given a value for the common parameter θ , this expression only depends on the parameter ρ . In other words,

$$\mathbf{E}_{m, r} \left[\left. \frac{\partial \log g_{T, t'}^{m, r}}{\partial \tau_t^{m \rightarrow B, r}} \right|_{t'=\bar{t}} \right] = f_T(\rho | \boldsymbol{\theta}, \mathbf{w}_{\bar{t}})$$

Hence, by mapping the average $\mathbf{E}_{m, r} \left[\left. \frac{\partial \log g_{T, t'}^{m, r}}{\partial \tau_t^{m \rightarrow B, r}} \right|_{t'=\bar{t}} \right]$ to a credibly estimated value of the average treatment effect, we can pin down ρ .

The remaining step is to spell out the conditions under which the empirical estimate of the gravity coefficient from Section 4 identifies the average treatment effect in Equation (D.24). To this end, we can express the error term $\varepsilon_{T, t}^{n, s}$ explicitly as

$$\begin{aligned} \varepsilon_{T, t}^{n, s} = & \left(\left. \frac{\partial \log g_{T, t'}^{n, s}}{\partial \tau_t^{n \rightarrow B, s}} \right|_{t'=\bar{t}} - \mathbf{E}_{m, r} \left. \frac{\partial \log g_{T, t'}^{m, r}}{\partial \tau_t^{m \rightarrow B, r}} \right|_{t'=\bar{t}} \right) \tau_t^{n \rightarrow B, s} + \\ & \log \lambda_t^{n, s} + \left(-\theta + \left. \frac{\partial \log g_{T, t'}^{n, s}}{\partial \log d_t^{n, s}} \right|_{t'=\bar{t}} \right) \log d_t^{n, s} + \left. \frac{\partial \log g_{T, t'}}{\partial \log \left(\frac{w_t^n}{w_t^B} \right)} \right|_{t'=\bar{t}} \log \left(\frac{w_t^n}{w_t^B} \right) - M_t, \end{aligned}$$

where M_t is an additive term introduced to ensure that $\varepsilon_{T, t}^{n, s}$ has mean zero for all t , with the corresponding adjustment absorbed by the fixed effects.

For the empirical estimate to identify the average treatment effect, we need that, conditional on the set of fixed effects, $\mathbf{E} \left[\varepsilon_{T, t}^{n, s} \tau_t^{n \rightarrow B, s} \right] = 0$. This requires two set of conditions. First (top line of the $\varepsilon_{T, t}^{n, s}$ expression), it requires that the (n, s) -specific responsiveness to tariffs $\left(\left. \frac{\partial \log g_{T, t'}}{\partial \tau_t^{n \rightarrow B, s}} \right|_{t'=\bar{t}} \right)$ is orthogonal to the square of tariffs (conditional independence is a stronger but sufficient condition). This is a common condition required for identification of the average treatment effect in settings

where, a priori, treatment effects are observation-specific. Second (bottom line of the $\varepsilon_{T,t}^{n,s}$ expression), it requires that tariffs are orthogonal to productivity ($\log \lambda_t^{n,s}$), transfer costs ($\log d_t^{n,s}$) and their interaction with the (n, s) -specific responsiveness ($\left. \frac{\partial \log g_{T,t'}^{n,s}}{\partial \log d_{t'}^{n,s}} \right|_{t'=\bar{t}}$), and relative wages interacted with the (n, s) -specific responsiveness ($\left. \frac{\partial \log g_{T,t'}^{n,s}}{\partial \log \left(\frac{w_t^n}{w_t^B} \right)} \right|_{t'=\bar{t}}$). The conditional quasi-random assignment of tariffs given by our IV strategy ensures that these conditions are satisfied in the empirical setting.

D.3.2 Derivation of Proposition 2 and interpretation of $f_{\text{cit}}(\cdot)$ and ε_{cit}

Recall that we assume an idea is patented and thus appears in the citation data (as citing its specific source) when its quality q' exceeds a sector- and time-specific threshold δ_t^s .

Using Equations (14) and (15) and the fact that $F_{k,t}^{n,s}(q)$ is distributed Fréchet, the total rate of arrival of new patentable ideas originating from (n, s, k) is

$$\mathcal{A}_{k,t}^{n,s} = \int_0^\infty A_{k,t}^{n,s} \left(\frac{\delta_t^s}{\zeta_t^{n,s} \omega_k(q)^\beta} \right) dF_{k,t}^{n,s}(q) \propto \alpha_t^s (\pi_{k,t}^{n \rightarrow B,s})^{1-\beta} \delta_t^s \lambda_t^{n,s} g_{k,t}^{n,s} (\zeta_t^{n,s} \omega_k)^\theta, \quad (\text{D.26})$$

where $\pi_{k,t}^{n \rightarrow B,s}$ is the share of sector- s varieties in Brazil sourced from (n, s, k) , and $g_{k,t}^{n,s}$ captures the correlation between productivity draws from imports and transfers, as defined in the previous subsection:

$$g_{T,t}^{n,s} \equiv \frac{\partial G_t^{B,s}}{\partial(n, T)} = \left[1 + \left(\frac{w_t^n \bar{\kappa}^{n \rightarrow B,s} (1 + \tau_t^{n \rightarrow B,s})}{w_t^B d_t^{n,s}} \right)^{\frac{-\theta}{1-\rho}} \right]^{-\rho},$$

$$g_{X,t}^{n,s} \equiv \frac{\partial G_t^{B,s}}{\partial(n, X)} = \left[1 + \left(\frac{w_t^B d_t^{n,s}}{w_t^n \bar{\kappa}^{n \rightarrow B,s} (1 + \tau_t^{n \rightarrow B,s})} \right)^{\frac{-\theta}{1-\rho}} \right]^{-\rho}.$$

From Equation (D.26), substituting the expressions for $\pi_{k,t}^{n \rightarrow B,s}$ (Equation D.12), summing across $k \in (X, T)$, and grouping all terms that only vary at the (s, t) level into a term denoted Z_t^s , the total number of citations to country n and sector s can be expressed as

$$\text{cit}_t^{n,s} \propto Z_t^s (\zeta_t^{n,s})^\theta (\lambda_t^{n,s})^{2-\beta} \sum_k (\tilde{w}_{k,t}^{n \rightarrow B} \bar{\kappa}_{k,t}^{n \rightarrow B,s})^{-\theta(1-\beta)} (g_{k,t}^{n,s})^{2-\beta} (\omega_k)^\theta. \quad (\text{D.27})$$

Taking logs of Equation (D.27) and taking a first-order approximation around a baseline period $t = \bar{t}$, for any arbitrary perturbation in productivity ($\lambda_t^{n,s}$), transfer costs ($d_t^{n,s}$), tariffs ($\tau_t^{n \rightarrow B,s}$),

and resulting changes in relative wages (w_t^n/w_t^B), we obtain the following expression:

$$\log \text{cit}_t^{n,s} \approx \mu_{n,s}^1 + \mu_{n,t}^2 + \mu_{s,t}^3 + \mathbf{E}_{m,r} \left[\sum_k \Omega_{k,t'}^{m,r} \eta_{k,t'}^{m,r} \Big|_{t'=\bar{t}} \right] \tau_t^{n \rightarrow B,s} + \varepsilon_{\text{cit},t}^{n,s}, \quad (\text{D.28})$$

where the term $\varepsilon_{\text{cit},t}^{n,s}$ has mean zero and all terms that only vary at the level of the country-sector, country-time, and sector-time, are grouped in their respective fixed effect, $\mu_{\cdot,\cdot}^i$.

Hence, the coefficient of the citations gravity is

$$\begin{aligned} \text{Structural coeff.} \\ (\text{citations gravity}) \end{aligned} = \mathbf{E}_{m,r} \left[\sum_k \Omega_{k,t'}^{m,r} \eta_{k,t'}^{m,r} \Big|_{t'=\bar{t}} \right], \quad (\text{D.29})$$

where

$$\Omega_{k,t}^{m,r} \equiv \frac{(\tilde{w}_{k,t}^{n \rightarrow B} \bar{\kappa}_{k,t}^{n \rightarrow B,s})^{-\theta(1-\beta)} (g_{k,t}^{n,s})^{2-\beta} (\omega_k)^\theta}{\sum_l (\tilde{w}_{l,t}^{n \rightarrow B} \bar{\kappa}_{l,t}^{n \rightarrow B,s})^{-\theta(1-\beta)} (g_{l,t}^{n,s})^{2-\beta} (\omega_l)^\theta} \quad \text{and} \quad \eta_{k,t}^{m,r} \equiv -\theta(1-\beta) \mathbf{1}_{\{k=X\}} + (2-\beta) \frac{\partial \log g_{k,t}^{m,r}}{\partial \tau_t^{m \rightarrow B,r}}.$$

Intuitively, the coefficient in Equation (D.28) is a weighted average of the semi-elasticities $\eta_{k,t}^{m,r}$ of learning through each channel $k = \{X, T\}$, where the weights $\Omega_{k,t}^{m,r}$ reflect the relative importance of each learning mode in total knowledge flows from (m, r) .

Using the expression for $\frac{\partial \log g_{T,t}^{m,r}}{\partial \tau_t^{m \rightarrow B,r}}$ derived in the previous subsection, and the analogous one for $\frac{\partial \log g_{X,t}^{m,r}}{\partial \tau_t^{m \rightarrow B,r}}$, we can compute the elasticities $\eta_{k,t}^{m,r}$ and their weights $\Omega_{k,t}^{m,r}$ in closed form as functions of ω_T , given values for ρ , θ , β , and observable or invertible quantities:

$$\mathbf{E}_{m,r} \left[\sum_k \Omega_{k,t'}^{m,r} \eta_{k,t'}^{m,r} \Big|_{t'=\bar{t}} \right] = f_{\text{cit}}(\omega_T | \rho, \boldsymbol{\theta}, \mathbf{w}_{\bar{t}}).$$

Also in this case, the error term $\varepsilon_{\text{cit},t}^{n,s}$ has zero mean by construction, and includes the (n, s) -specific responses to changes in wages and transfer costs, productivity and learning efficiency ($\log \zeta_t^{n,s}$), and the difference between the (n, s) -specific and average responsiveness to tariffs. The condition for identification of the average treatment effect is that, conditional on the set of fixed effects, $\mathbf{E} \left[\varepsilon_{\text{cit},t}^{n,s} \tau_t^{n \rightarrow B,s} \right] = 0$. This condition holds by construction in the model, and that is provided by the IV strategy in the empirical analysis.

D.3.3 Computing $f_T(\cdot)$ and $f_{\text{cit}}(\cdot)$ in the quantification

To calibrate ρ and ω_T , we evaluate Equations (D.24) and (D.29) for all 13 3-digit ISIC sectors used in the quantitative analysis for which there is a positive number of technology transfers in the initial BGP. We then compute the averages of these sector-specific coefficients, and update ρ and ω_T until the resulting averages are equal to the estimated coefficients (bottom panel of Table 7).

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